AnyFusion: Train Once, Fuse Anywhere—Universal Unsupervised Hyperspectral–Multispectral Image Fusion via Wavelength-Aware Spectral Set Transformer

Ke Zheng, Qianqian Zhao, Lianru Gao, *Senior Member, IEEE*, Haoyang Yu, *Senior Member, IEEE*, Jiaxin Li, *Member, IEEE*, and Bing Zhang, *Fellow, IEEE*

***Abstract*—Fusing a low-resolution hyperspectral image (LrHSI) with a high-resolution multispectral image (HrMSI) is a practical route to high-resolution hyperspectral imagery (HrHSI), yet real deployment faces three obstacles: no HrHSI supervision exists for a new sensor; sensors differ arbitrarily in band count, wavelength layout, spectral response function (SRF), and resolution ratio; and existing unsupervised methods reoptimize from scratch for every scene, so nothing carries over between images and, with no reference available, an unstable run cannot even be detected. We propose AnyFusion, a universal unsupervised fusion framework that is trained once and then adapted, without any HrHSI supervision, to arbitrary sensor configurations. AnyFusion represents every band as a token conditioned on its physical center wavelength and processes the band set with a spectral set Transformer (SST) whose attention is guided by wavelength and SRF-overlap biases; a wavelength-query decoder synthesizes arbitrary output bands at arbitrary spatial scales. The unknown degradation is estimated per scene by jointly learning an implicit neural point spread function (PSF) and a wavelength-conditional SRF; the learned operators match the fusion accuracy of the ground-truth operators without requiring knowledge of the kernel or of the sampling geometry. On this band-agnostic architecture, a cross-sensor universal pretraining stage driven purely by consistency losses yields a single checkpoint that transfers to unseen sensors. On four benchmarks, AnyFusion attains the best mean peak signal-to-noise ratio (PSNR) among all compared unsupervised methods on three datasets, consistent pretraining gains of +0.62 to +2.77 dB with 2.0–3.3× fewer adaptation iterations at matched from-scratch (FS) quality, and strong generalization across spatial scales, band subsets, wavelength ranges, and SRFs. On a real ZY1-02D spaceborne pair with no reference, a checkpoint pretrained solely on the simulated benchmarks raises spectral observation consistency by +5.18 dB (+2.79 dB when all runs are scored with the sensor-nominal SRF) while converging roughly an order of magnitude faster, supporting "train once, fuse anywhere."**

***Index Terms*—Cross-sensor generalization, hyperspectral–multispectral fusion, spectral set Transformer (SST), universal pretraining, unsupervised learning, wavelength embedding.**

Received 17 August 2026; revised 27 September 2026; accepted 14 October 2026. Date of publication 22 October 2026; date of current version 30 October 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42325104, Grant 42671613, Grant 42471380, and Grant 42201362; in part by Shandong Provincial Natural Science Foundation of China under Grant ZR2024QD042; in part by the Science and Technology Research Program of Chongqing Municipal Education Commission under Grant KJQN202620606; in part by the Postdoctoral Fellowship Program (Grade B) of China Postdoctoral Science Foundation under Grant GZB20260443; and in part by Chongqing Postdoctoral Science Foundation under Grant 2025CQBSHTB2012. *(Corresponding author: Lianru Gao.)*

Ke Zheng and Qianqian Zhao are with the College of Geography and Environment, Liaocheng University, Liaocheng 252059, China (e-mail: zhengke@lcu.edu.cn; 2510140243@stu.lcu.edu.cn).

Lianru Gao is with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: gaolr@aircas.ac.cn).

Haoyang Yu is with the Information Science and Technology College, Dalian Maritime University, Dalian 116026, China (e-mail: yuhy@dlmu.edu.cn).

Jiaxin Li is with the School of Computer Science and Technology (National Exemplary Software School), Chongqing University of Posts and Telecommunications, Chongqing 400065, China (e-mail: lijiaxin203@mails.ucas.ac.cn).

Bing Zhang is with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: zb@radi.ac.cn).

Data is available online at https://github.com/saber-zero/AnyFusion

Digital Object Identifier 10.1109/TGRS.2026.3745681

NOMENCLATURE

$\mathbf{X}, \hat{\mathbf{X}}$	Latent/reconstructed HrHSI, $\mathbb{R}^{C\times H\times W}$.
$\mathbf{Y}$	Observed LrHSI, $\mathbb{R}^{C\times h\times w}$.
$\mathbf{Z}$	Observed HrMSI, $\mathbb{R}^{M\times H\times W}$.
C, M	Number of HSI/MSI bands.
λ_c, ν_m	Center wavelengths of HSI band c/MSI band m.
s	Spatial scale factor ($h = H/s$, $w = W/s$).
k, $\mathbf{R}$	PSF kernel and SRF matrix ($\mathbb{R}^{M\times C}$).
$\gamma(\cdot)$	Fourier positional encoding.
$\mathbf{e}_n$	Conditioning vector of band token n, $\mathbb{R}^D$.
$\mathbf{T}_n$	Band token n, $\mathbb{R}^{d_b\times H\times W}$.
N	Number of band tokens ($N = C + M$).
D, d_b	Hidden dimension/per-band token dimension.
n_b, n_h	Number of SST blocks/attention heads.
$\mathbf{F}$	Fused feature map, $\mathbb{R}^{D\times H\times W}$.

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) record the radiance of a scene in hundreds of narrow, contiguous spectral bands and thereby enable fine-grained material discrimination in applications such as precision agriculture [1], mineral mapping [2], forest monitoring [3], wetland vegetation mapping [4], and

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

anomaly detection [5]. However, because the photon energy collected by a sensor must be divided among its spectral channels, hyperspectral instruments inevitably trade spatial resolution for spectral resolution, and spaceborne or airborne HSIs are typically delivered at a coarse ground sampling distance. Multispectral images (MSIs), in contrast, offer a much finer spatial resolution but only a handful of broad bands. Fusing a low-resolution HSI (LrHSI) with a high-resolution MSI (HrMSI) of the same scene, so as to reconstruct a high-resolution HSI (HrHSI), has therefore become a central topic in remote sensing image processing [6], [7], [8], [9].

Two decades of research have produced a broad range of fusion algorithms. Model-based methods formulate fusion as an inverse problem regularized by handcrafted priors, including pansharpening extensions [10], [11], [12], Bayesian estimators [13], [14], [15], spectral unmixing and matrix factorization [16], [17], [18], and tensor decomposition [19], [20], [21]. More recently, supervised deep networks trained on simulated LrHSI–HrMSI–HrHSI triplets under the Wald protocol [22] have improved fusion accuracy, evolving from convolutional architectures [23], [24], [25] to attention- and Transformer-based designs [26], [27], [28], state-space models [29], diffusion models [30], and implicit neural fusion functions [31].

Despite this progress, deploying a fusion network on a real mission exposes a fundamental mismatch. Supervised training requires HrHSI targets, but for a newly launched or newly paired sensor no such reference exists, since the HrHSI is precisely the quantity to be estimated. Unsupervised per-image methods circumvent this by optimizing a network on a single LrHSI–HrMSI pair using degradation-consistency objectives, often while estimating the unknown point spread function (PSF) and spectral response function (SRF) [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42]. These methods are supervision-free and degradation-adaptive, but they pay for this flexibility twice. First, every scene is fused by training a network from scratch and everything learned on previous scenes is discarded, so the full optimization cost recurs for every new image and no knowledge ever accumulates across a mission. Second, each fusion is an independent nonconvex optimization from a random initialization, and because no reference exists at deployment, a poorly converged run cannot even be detected, let alone corrected.

A second, subtler obstacle is configuration heterogeneity. Existing fusion networks hard-wire the sensor configuration into their architecture: the first convolution fixes the number of input bands, the last layer fixes the number of output bands, and upsampling modules fix the resolution ratio. A model trained for one HSI–MSI pair is thus structurally incompatible with a sensor that has a different band count, wavelength coverage, SRF, or scale factor. Recent work has begun to address band-number variability by slicing a large shared kernel according to the channel index [43], but a channel index carries no physical meaning: band 30 of one sensor and band 30 of another may lie hundreds of nanometers apart. Such index-based encodings cannot express which target wavelengths are physically observable from the inputs, and they still require supervised fine-tuning on the new sensor.

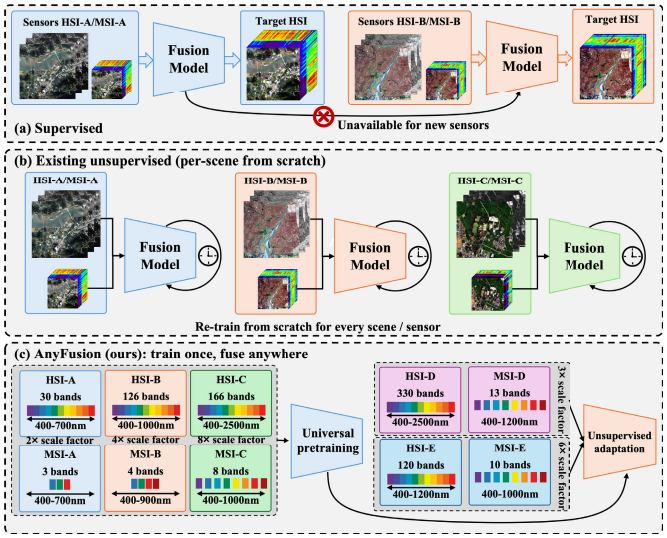


Fig. 1. Comparison of fusion deployment paradigms. (Top) Supervised methods require HrHSI labels unavailable for new sensors; (middle) existing unsupervised methods retrain a sensor-specific network from scratch for every scene; and (bottom) AnyFusion pretrains one band-agnostic network across heterogeneous sensors with purely unsupervised losses, then adapts it to any new sensor configuration (band count, wavelength layout, SRF, and scale) with a short unsupervised adaptation.

What is missing is an architecture whose spectral reasoning is anchored to the *physical wavelength axis* shared by all optical sensors, combined with a training paradigm that accumulates knowledge across sensors *without* HrHSI supervision, so that per-scene optimization starts from a shared representation rather than from scratch, and its cost and run-to-run variability shrink accordingly.

To close this gap, we propose *AnyFusion*, a universal unsupervised HSI–MSI fusion framework built on the principle of "train once, fuse anywhere" (see Fig. 1). AnyFusion treats every spectral band, whether hyperspectral or multispectral, as a set element conditioned on its physical center wavelength. A *spectral set Transformer* (SST) backbone keeps per-band tokens throughout the network and alternates spectral set attention, biased by continuous interband wavelength differences and by the physical SRF overlap between MSI and HSI bands, with wavelength-conditioned spatial mixing; a wavelength-query decoder then reconstructs an arbitrary set of target bands at an arbitrary spatial scale. Because no layer depends on the band count, wavelength layout, SRF, or scale, one set of weights applies to any sensor configuration. The unknown degradation of a specific scene is recovered by *jointly* learning an implicit neural representation (INR) of the PSF and a wavelength-conditional SRF from the observations alone; beyond enabling unsupervised training, we show that the learned degradation matches the fusion accuracy of the ground-truth operators without knowledge of the kernel or of the sampling geometry, to which fixed operators are highly sensitive. Finally, a *universal pretraining* stage optimizes the band-agnostic backbone across multiple heterogeneous datasets using purely unsupervised consistency losses. This yields a single checkpoint that adapts to an unseen sensor in a small fraction of the from-scratch (FS) iterations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SUMMARY OF THE FOUR BENCHMARK DATASETS USED IN THE EXPERIMENTS. ALL DATASETS ARE NORMALIZED TO [0, 1]. THE SRF FOR GENERATING HRMSI IS UNIFORMLY SET TO S2B

Dataset	Size	Bands	Range (nm)	MSI	Sensor
PaviaU	336×336	103	430–838	8	ROSIS
Houston18	240×240	46	389–1033	10	CASI
Chikusei	512×512	128	363–1018	10	Headwall
TG	240×240	54	413–888	9	Tiangong

Retained S2B bands: PaviaU B1–B8; Houston18 and Chikusei B1–B8, B8A, B9; TG B1–B8, B8A. A band is retained if its response, sampled at the HSI band centers, is nonzero; each retained response is renormalized over the covered range (e.g., only 58% of the B8 response area lies within PaviaU's range).

The main contributions of this article are summarized as follows.

1) We propose a universal unsupervised fusion paradigm that combines cross-sensor unsupervised pretraining with rapid per-image unsupervised adaptation. Pretraining improves the final accuracy on all four benchmarks (+0.62 to +2.77 dB in three-seed mean, with the seed std reduced by 4–26×) while matching FS final quality with 2.0–3.3× fewer adaptation iterations, and no HrHSI ever enters the training gradients (Sections IV-B and IV-C).
2) We design a wavelength-aware SST that is agnostic to the number, layout, and modality of input bands. Spectral attention is grounded in physics through a continuous $\Delta\lambda$ relative bias and an SRF-overlap routing prior, and a wavelength-query decoder supports arbitrary output band sets and spatial scales. A single-variable ablation shows that anchoring the band embedding to physical wavelength, rather than channel index, is what makes cross-sensor pretraining transfer (Section IV-D).
3) We introduce a joint degradation-learning stage that couples an implicit-neural-representation PSF with a wavelength-conditional SRF. The wavelength-continuous SRF parameterization extrapolates consistently to unobserved wavelengths, and the jointly learned operators reach oracle-level fusion accuracy without knowledge of the kernel or of the sampling geometry, and they remain robust under markedly non-Gaussian blur (Section IV-F).
4) We provide evidence for cross-configuration generalization: pretrained transfer across spatial scales (2×–6×) and different SRFs (Sentinel-2B (S2B) and Landsat-8), and architectural flexibility across band subsets and disjoint wavelength ranges, including a physically interpretable failure boundary when target wavelengths are not covered by any input band (Section IV-D). Beyond simulation, we validate the complete pipeline on a real ZY1-02D spaceborne HSI–MSI pair with no reference, where a checkpoint pretrained only on the simulated benchmarks transfers to this unseen sensor and improves spectral observation consistency by +5.18 dB in three-run mean (+2.79 dB under the sensor-nominal SRF; Section IV-E).

The remainder of this article is organized as follows. Section II reviews related work. Section III details the proposed AnyFusion framework. Section IV reports experimental results and analyses on four benchmark datasets and a real ZY1-02D pair. Section V concludes the article.

II. RELATED WORK

A. Model-Based HSI–MSI Fusion

Model-based methods cast fusion as an ill-posed inverse problem defined by an explicit observation model, in which the LrHSI is a blurred and downsampled version of the latent HrHSI and the HrMSI is its spectral projection through the SRF. Early studies extended pansharpening techniques such as component substitution and multiresolution analysis to the hyperspectral case [10], [11], [12], while Bayesian formulations imposed statistical priors on the latent image [13], [14], [15], [44]. Spectral-unmixing approaches, exemplified by coupled nonnegative matrix factorization (CNMF) [16] and sparse representation variants [17], exploit the low-rank structure of hyperspectral data by alternately estimating endmembers and abundances from the two observations; subspace-regularized [18] and tensor-decomposition [19], [20], [21] formulations further enrich this family. These methods require no training data, but their performance hinges on the accuracy of handcrafted priors, and most of them assume the PSF and SRF to be known a priori, an assumption rarely satisfied in cross-sensor deployment [6].

B. Deep Learning-Based Fusion

Supervised deep networks learn the fusion mapping from simulated triplets generated under the Wald protocol [22]. Representative designs include model-guided unfolding networks [23], [45], spatial–spectral attention convolutional neural networks (CNNs) [24], [25], cross-fusion Transformers [26], [27], lightweight adversarial models [28], state-space architectures [29], and conditional diffusion models [30]. FeINFN [31] introduced an implicit neural fusion function in both spatial and frequency domains, showing that continuous representations are effective for fusion. Beyond two-source fusion, gradual frameworks fuse HSI, MSI, and panchromatic images and show that the fused products benefit downstream vegetation parameter retrieval [46]. Despite their accuracy, these methods presuppose HrHSI training labels and inherit the sensor configuration of the training set, so they cannot be applied directly to a new sensor without paired data and architectural surgery.

Unsupervised methods remove the need for HrHSI labels by optimizing on a single observation pair. The unmixing line couples two autoencoders that share abundances, from the sparse Dirichlet network uSDN [32] to CUCaNet [34], which adds cross-attention and enforces consistency between the two modality branches. A parallel line integrates degradation learning: the coupled CNN of Zheng et al. [33] learns adaptive response functions jointly with fusion, and blind super-resolution methods estimate the PSF/SRF alternately with the latent image [35]; FusionNet [47] casts unsupervised fusion as convolutional variational inference, UDALN [48] couples blind degradation estimation with a lightweight fusion network, and a hybrid Transformer–CNN network learns

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

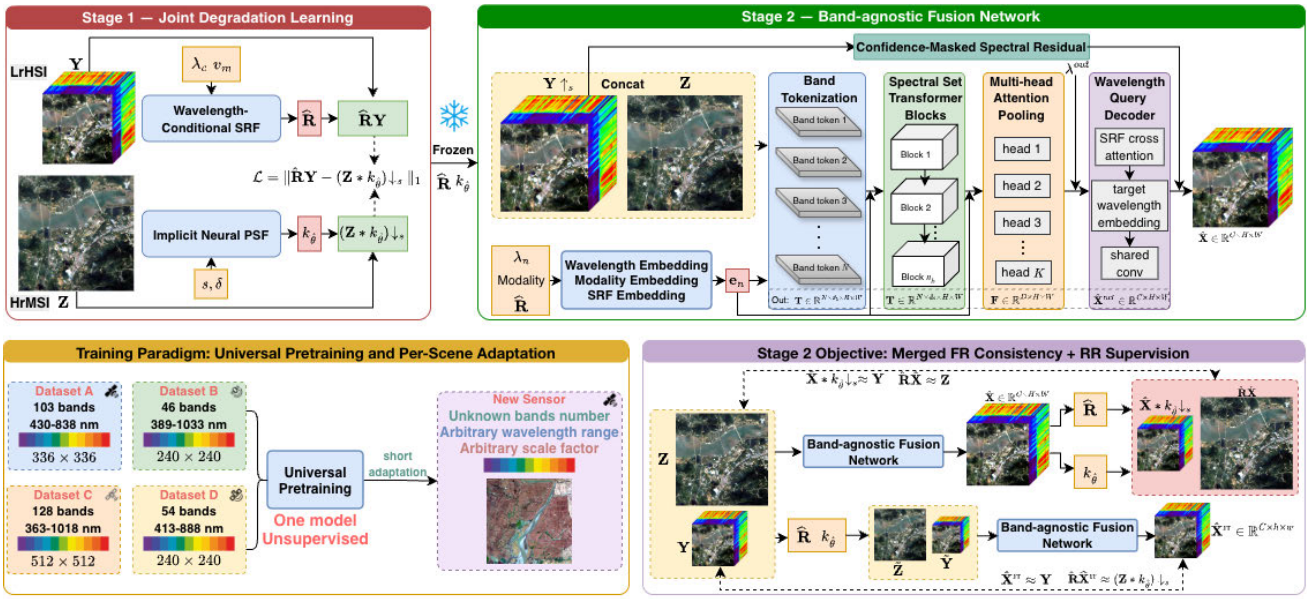


Fig. 2. Overall framework of AnyFusion. **Stage 1** jointly learns the degradation from the observations alone: an implicit neural PSF $k_{\hat{\theta}}$ and a wavelength-conditional SRF $\hat{\mathbf{R}}(\lambda)$ are optimized by cross-consistency between the spectrally degraded LrHSI and the spatially degraded HrMSI. **Stage 2** optimizes the band-agnostic fusion network under the frozen learned degradation with a merged objective: a full-resolution (FR) consistency branch on $(\mathbf{Y}, \mathbf{Z})$ and a reduced-resolution (RR) supervised branch on observations redegraded by $(k_{\hat{\theta}}, \hat{\mathbf{R}})$. The fusion network lifts every HSI/MSI band into a wavelength-conditioned token, processes the token set with n_b SST blocks, pools the tokens with multihead attention, and decodes arbitrary target bands via wavelength queries. A universal pretraining stage (bottom) optimizes the same network across heterogeneous datasets with purely unsupervised consistency losses before per-scene adaptation.

degradation representations for blind unsupervised fusion [49]. Model-inspired designs embed the observation model into the network itself: MIAE [36] derives an autoencoder whose decoder mirrors the degradation operators, achieving strong accuracy with a compact model; subsequent work refined this direction with mutual cross-modality learning [37], attention-embedded degradation learning [39], multistage model-informed optimization [40], model-guided coarse-to-fine fusion [38], deep spectral–spatial collaborative constraints [41], deep Tucker decomposition with manifold learning [50], coupled spatial–spectral deep image priors (DIPs) [51], enhanced DIPs [42], and spectral diffusion priors [52]. Test-time training [53] and semi-supervised multistage schemes [54] blur the boundary between supervised and unsupervised regimes. Two limitations persist across this literature. First, each scene is optimized from scratch from a random initialization, so the full optimization cost recurs for every image, no knowledge accumulates across scenes, and the outcome varies from run to run with no reference to flag a poor one. Second, the architectures index spectral bands by channel position, binding the learned weights to one specific sensor configuration. AnyFusion addresses both: its band-agnostic backbone is pretrained once across sensors and then adapted per scene in a fraction of the FS iterations.

C. Cross-Sensor Generalization and Universal Pretraining

Large-scale pretraining has recently reshaped remote sensing, with masked-autoencoding foundation models learning transferable representations from unlabeled imagery [55], [56]. Recent foundation models make the representation itself sensor-aware by conditioning on spectral metadata, e.g., hypernetworks driven by band wavelengths (DOFA [57]) or by wavelength, bandwidth, and acquisition metadata (Copernicus-FM [58]), wavelength-based channel cross-attention (Panopticon [59]), full-SRF and ground-sampling-distance encodings (SenPa-MAE [60]), and spectrum-aware projections (SMARTIES [61]). They learn encoders for downstream recognition and reconstruct at most masked patches of their input bands; none synthesizes bands absent from its input or performs spatial–spectral reconstruction under a physical observation model; AnyFusion performs this reconstruction with a wavelength- and SRF-conditioned generative network trained by label-free consistency. Within fusion, pretraining-guided frameworks [62], [63] demonstrate that representations learned on source scenes accelerate fusion of target scenes, but they operate on fixed band layouts. To handle variable band numbers, the recent spectral-band and fusion-scale agnostic (SSA) framework [43] slices a shared nested (Matryoshka) convolution kernel according to the channel index, an encoding that ignores the physical wavelength of each band and therefore cannot generalize across sensors whose channels sample different spectral positions. In the adjacent field of spectral reconstruction from RGB, conditioning on the camera spectral sensitivity has proven valuable [64], [65], [66], suggesting that response functions are informative conditioning signals. Meanwhile, INRs [67], [68] with Fourier feature encodings [69] and local implicit image functions [70] show that continuous coordinate conditioning supports arbitrary-resolution querying, and the spatial–spectral implicit function (SSIF) extends such implicit functions jointly to continuous wavelengths for supervised single-image spatial–spectral super-resolution [71]. AnyFusion combines these ideas for unsupervised fusion: bands are embedded by their physical wavelengths, the MSI SRF conditions both attention and decoding, an implicit neural

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

PSF and a wavelength-conditional SRF model the degradation continuously, and set-style attention [72], [73], [74] makes the whole network invariant to the size and ordering of the band set. To our knowledge, this yields the first train-once, fuse-anywhere paradigm for unsupervised HSI–MSI fusion.

III. Methodology

A. Problem Formulation and Overview

Let $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ denote the latent HrHSI with C spectral bands centered at wavelengths $\{\lambda_c\}_{c=1}^{C}$, and let the two observations be the LrHSI $\mathbf{Y} \in \mathbb{R}^{C\times h\times w}$ and the HrMSI $\mathbf{Z} \in \mathbb{R}^{M\times H\times W}$ with $M \ll C$ bands centered at $\{\nu_m\}_{m=1}^{M}$, where $h = H/s$ and $w = W/s$ for a spatial scale factor s. The standard observation model relates them through two degradation operators

$$\mathbf{Y} = (\mathbf{X} * k)\downarrow_s, \quad \mathbf{Z} = \mathbf{R}\mathbf{X} \tag{1}$$

where k is the PSF blur kernel, assumed spectrally invariant (i.e., shared by all bands) as in the standard observation model and in our simulation protocol, $\downarrow_s$ denotes downsampling by factor s, and $\mathbf{R} \in \mathbb{R}^{M\times C}$ is the SRF matrix that projects the C hyperspectral bands onto the M multispectral bands. Throughout this article, matrices and tensors are denoted by bold uppercase letters, vectors by bold lowercase letters, and scalars by italic letters; $\hat{\cdot}$ marks estimated quantities. Nomenclature summarizes the main notation.

We target the *universal unsupervised* setting: neither $\mathbf{X}$, k, nor $\mathbf{R}$ is available during training, and the sensor configuration (C, M, $\{\lambda_c\}$, $\{\nu_m\}$, the SRF shape, and the scale s) varies from deployment to deployment. The goal is a single network f_Θ that, given any $(\mathbf{Y}, \mathbf{Z})$ pair together with its wavelength metadata, reconstructs $\hat{\mathbf{X}}$ after a short unsupervised adaptation.

Terminology: Here "unsupervised" means that no HrHSI serves as a training target at any stage, for the target scene or for the pretraining pool; "pretraining" is a one-time, label-free optimization of the same consistency objective on LrHSI–HrMSI pairs of other sensors, which provides an initialization rather than supervision.

As illustrated in Fig. 2, AnyFusion couples a band-agnostic fusion network with a two-stage optimization pipeline. *Stage 1* (see Section III-B) estimates the scene-specific degradation by jointly optimizing an implicit neural PSF and a wavelength-conditional SRF from the observations alone. *Stage 2* then optimizes the band-agnostic fusion network under the frozen learned degradation, using a merged objective that couples FR consistency with RR supervision synthesized by the learned operators (see Section III-E). The fusion network itself (see Sections III-C and III-D) is a wavelength-aware SST followed by a wavelength-query decoder; because every module is conditioned on physical wavelengths rather than channel indices, the same weights apply to any sensor configuration, which in turn enables cross-sensor universal pretraining (see Section III-E).

B. Stage 1: Joint Degradation Learning

Unsupervised fusion relies on consistency with the observation model (1), so an estimate of $(k, \mathbf{R})$ that is consistent with the observations is a prerequisite; as Section IV-F will show, consistency must cover both the kernel shape and the sampling geometry of the data. In cross-sensor deployment these operators are unknown and cannot be assumed Gaussian or box-shaped. We therefore parameterize both operators as *continuous functions* of the spatial offset for the PSF and of the physical wavelength for the SRF, so that they remain well-defined for any scale factor and any wavelength subset, which is essential for the cross-configuration experiments in Section IV. The ablation in Section IV-F verifies that this parameterization remains robust under markedly non-Gaussian blur (motion, anisotropic, and bimodal PSFs), where joint learning stays within about 1 dB of the exact ground-truth kernel.

1) Implicit Neural PSF: The PSF is modeled as an INR over the continuous spatial offset $\boldsymbol{\delta} = (\delta_x, \delta_y)$ expressed in LrHSI pixel units. A multilayer perceptron (MLP) predicts a residual on top of a Gaussian log-prior, and a softmax over the sampled offsets guarantees a normalized, nonnegative kernel

$$k_{\hat{\theta}}(\boldsymbol{\delta}) = *softmax_{\boldsymbol{\delta}}\left(-\frac{\|\boldsymbol{\delta}\|^2}{2\sigma_0^2} + \mathrm{MLP}_{\hat{\theta}}(\boldsymbol{\gamma}(\boldsymbol{\delta})) + \log\mu(\boldsymbol{\delta})\right) \tag{2}$$

where $\boldsymbol{\gamma}(\cdot)$ is a Fourier positional encoding, σ_0 sets the Gaussian prior width, and $\mu(\boldsymbol{\delta}) = \mathrm{sigmoid}(-\kappa_\mu(\|\boldsymbol{\delta}\| - r_s))$ is a soft radial mask with support radius $r_s = 1.5$ low-resolution (LR) pixels and sharpness $\kappa_\mu = 5$ that bounds the kernel support (its logarithm is floored at 10^{-8}). Since the representation lives in scale-invariant LrHSI offset coordinates, the discrete kernel for any (possibly fractional) scale s is obtained simply by sampling $k_{\hat{\theta}}$ on the corresponding $K \times K$ HrHSI-resolution grid of offsets, $K = 2\lceil r_s s\rceil + 1$, with the softmax taken over these K^2 samples. The MLP is zero-initialized, so optimization starts exactly from the Gaussian prior.

2) Wavelength-Conditional SRF: Rather than estimating a fixed $M \times C$ matrix tied to one band layout, we model the SRF as a continuous function of wavelength. An MLP maps the encoded wavelength to M response logits, again as a residual on a Gaussian prior centered at the MSI band centers

$$\hat{R}_m(\lambda) = *softmax_{\lambda\in\Lambda}\left(-\frac{(\lambda-\nu_m)^2}{2\beta^2} + \mathrm{MLP}_{\hat{\phi}}(\boldsymbol{\gamma}(\lambda))_m\right) \tag{3}$$

where β is the prior bandwidth and the softmax normalizes each MSI band's response over the queried wavelength set Λ; in practice Λ is the set of wavelengths at which the operator is applied (the input HSI band centers during Stage 1, and the full target band set when evaluating the spectral consistency of Stage 2). Because $\hat{R}_m(\cdot)$ is defined for every wavelength, an SRF fit on a subset of HSI bands evaluates consistently on the full band set, a property that later allows the network to be trained with subsampled inputs yet reconstruct all target bands.

3) Cross-Consistency Objective: Neither degraded observation of $\mathbf{X}$ is available in isolation, but degrading $\mathbf{Y}$ spectrally and $\mathbf{Z}$ spatially must yield the same LR MSI (see Fig. 3). Stage 1 therefore minimizes

$$\mathcal{L}_{\mathrm{S1}} = \left\|\hat{\mathbf{R}}\mathbf{Y} - \left(\mathbf{Z} * k_{\hat{\theta}}\right)\downarrow_s\right\|_1 + \eta_k\,\Omega_k + \eta_r\,\Omega_r \tag{4}$$

where the two regularizers encourage smooth, physically plausible operators by penalizing the first-order gradient of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

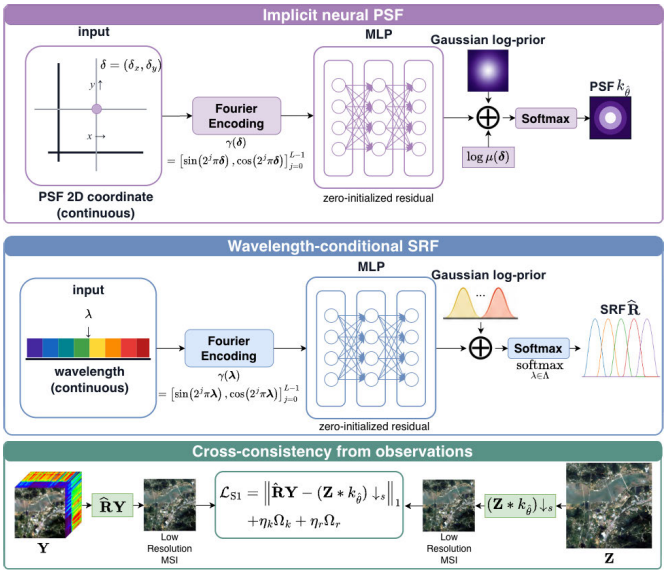


Fig. 3. Joint degradation learning in Stage 1. The implicit neural PSF $k_{\hat{\theta}}$ maps a continuous spatial offset (Fourier-encoded) to a kernel weight as a residual on a Gaussian prior, and can be sampled on the grid of any scale factor s. The wavelength-conditional SRF $\hat{\mathbf{R}}$ maps any wavelength λ to M response values as a residual on Gaussian priors centered at the MSI band centers ν_m. Both are optimized by the cross-consistency objective in (4): spectrally degrading the LrHSI ($\hat{\mathbf{R}}\mathbf{Y}$) and spatially degrading the HrMSI ($(\mathbf{Z} * k_{\hat{\theta}})\downarrow_s$) must produce the same LR MSI.

PSF representation and the second-order difference of the SRF logits along wavelength

$$\Omega_k = \tfrac{1}{2}\,\mathbb{E}_{\boldsymbol{\delta}} \left\| \nabla_{\boldsymbol{\delta}}\, \ell_{\hat{\theta}}\,(\boldsymbol{\delta}) \right\|_2^2$$

$$\Omega_r = \frac{1}{M\,(L-2)} \sum_{m=1}^{M} \sum_{i=2}^{L-1} \left(\ell^r_{m,i+1} - 2\ell^r_{m,i} + \ell^r_{m,i-1}\right)^2 .$$

Here, $\ell_{\hat{\theta}}$ and $\ell^r_{m,i}$ are the softmax arguments of (2) and (3); the expectation is estimated from 128 offsets drawn uniformly from $[-2r_s, 2r_s]^2$ per step, and the L sorted wavelengths are the input HSI band centers plus 64 wavelengths drawn uniformly from the HSI range extended by 30 nm. After Stage 1, both operators are frozen.

Because both operators are estimated from the observations themselves, they also absorb aspects of the acquisition geometry that a nominal operator does not encode. In particular, the relative sampling phase of the LrHSI and HrMSI grids is rarely known to subpixel accuracy; the INR PSF can place its mass anywhere within its support and thus recovers such an offset automatically. Section IV-F shows that the learned operators match the fusion accuracy of phase-matched ground-truth operators, whereas the same exact operators applied with a subpixel phase offset lose about 2 dB.

C. Wavelength-Aware SST

The SST backbone is built on one principle: a spectral band is a *set element* whose identity is its physical wavelength, not its channel index. The bicubic-upsampled LrHSI and the HrMSI are concatenated along the band axis into $N = C + M$ single-channel images, and every band is lifted into a token that traverses the entire network; bands are never collapsed into a fixed-size feature map until the final pooling. All operators below are permutation-equivariant along the band axis and independent of N, making the backbone band-agnostic by construction. Fig. 4 details the block design.

1) Wavelength-Conditioned Band Tokenization: Each band n carries a conditioning vector $\mathbf{e}_n \in \mathbb{R}^D$ that encodes its physical identity. The center wavelength is embedded by a Fourier encoding followed by an MLP:

$$\mathbf{e}^{\lambda}_n = \mathrm{MLP}\left(\boldsymbol{\gamma}\left(\lambda_n\right)\right) \tag{5}$$

where $\boldsymbol{\gamma}(\lambda_n) = [\bar{\lambda}_n, \{\sin(2^t\pi\bar{\lambda}_n), \cos(2^t\pi\bar{\lambda}_n)\}_{t=0}^{7}]$ are Fourier features of the center wavelength expressed in micrometers ($\bar{\lambda}_n$); throughout this section, λ_n denotes the center wavelength of token n, i.e., λ_c for an HSI token and ν_m for an MSI token. The backbone, the SRF embedding below, and the decoder of Section III-D each use their own copy of this embedding network (identical architecture, separate parameters). The full conditioning vector adds two further identity cues

$$\mathbf{e}_n = \mathbf{e}^{\lambda}_n + \mathbf{e}^{\mathrm{mod}}_n + \mathbb{I}\left[n \in \mathcal{M}\right]\, \tilde{\mathbf{e}}^{\mathrm{srf}}_{m(n)} + \mathbf{p}^{\mathrm{srf}} \tag{6}$$

where $\mathbf{e}^{\mathrm{mod}}_n \in \{\mathbf{e}^{\mathrm{HSI}}, \mathbf{e}^{\mathrm{MSI}}\}$ is a learned modality embedding, $\mathcal{M}$ is the set of MSI tokens, and $\tilde{\mathbf{e}}^{\mathrm{srf}}_m = \mathrm{MLP}\left(\sum_g \bar{R}_m(\lambda_g)\, \mathbf{e}^{\lambda}(\lambda_g)\right)$ embeds the area-normalized spectral response $\bar{R}_m$ of MSI band $m(n) = m$ (the Stage-1 estimate $\hat{\mathbf{R}}$, sampled at the target HSI band centers $\{\lambda_g\}$), as a response-weighted average of wavelength embeddings, so that broadband MSI channels are distinguishable from narrowband HSI channels sharing the same center. The global sensor prompt $\mathbf{p}^{\mathrm{srf}} = \mathrm{MLP}\left(1/M \sum_m \tilde{\mathbf{e}}^{\mathrm{srf}}_m\right)$, added to every token, informs all bands (HSI ones included) of which MSI sensor observes the scene. A shared two-layer convolution lifts each single-band image $\mathbf{x}_n \in \mathbb{R}^{1\times H\times W}$ to a d_b-channel token, modulated by feature-wise linear modulation (FiLM) [75]

$$\mathbf{T}_n = (1 + \mathbf{g}_n) \odot \mathrm{Conv}\left(\mathbf{x}_n\right) + \mathbf{b}_n, \quad \left[\mathbf{g}_n, \mathbf{b}_n\right] = \mathbf{W}_{\mathrm{FiLM}}\, \mathbf{e}_n \tag{7}$$

where $\mathbf{g}_n$ and $\mathbf{b}_n$ are the FiLM gain and bias and $\odot$ denotes channel-wise multiplication.

2) Spectral Set Attention With Physical Biases: Within each SST block, information is exchanged along the band axis by set attention [74]. So that the cost of the $N \times N$ attention logits does not grow with the spatial size, queries and keys are computed from a compact global descriptor of each token rather than from its full feature map: the token is average-pooled to a $p \times p$ window, flattened, and fused with its conditioning vector

$$\mathbf{f}_n = \mathbf{W}_{\mathrm{d}}\, \mathrm{vec}\left(\mathrm{P}_p\left(\mathrm{GN}\left(\mathbf{T}_n\right)\right)\right) + \mathbf{W}_{\mathrm{c}}\, \mathbf{e}_n \tag{8}$$

where $\mathrm{GN}(\cdot)$ denotes group normalization and $\mathrm{P}_p(\cdot)$ adaptive average pooling to $p \times p$; per-head queries and keys are the linear projections $\mathbf{q}_i = \mathbf{W}_{\mathrm{q}}\mathbf{f}_i$ and $\mathbf{k}_j = \mathbf{W}_{\mathrm{k}}\mathbf{f}_j$. The logits of each attention head $g \in \{1, \ldots, n_h\}$ are augmented with two physically grounded biases

$$A^{(g)}_{ij} = *softmax_j\left(\frac{\mathbf{q}_i^T \mathbf{k}_j}{\sqrt{d}} + b^{(g)}_{\Delta\lambda}\left(\lambda_i - \lambda_j\right) + \alpha^{(g)}\, B^{\mathrm{srf}}_{ij}\right) \tag{9}$$

where $b^{(g)}_{\Delta\lambda}(\lambda_i - \lambda_j) = [\mathrm{MLP}_{\Delta\lambda}(\lambda_i - \lambda_j)]_g$ is a small MLP of the continuous wavelength difference with one output per head,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

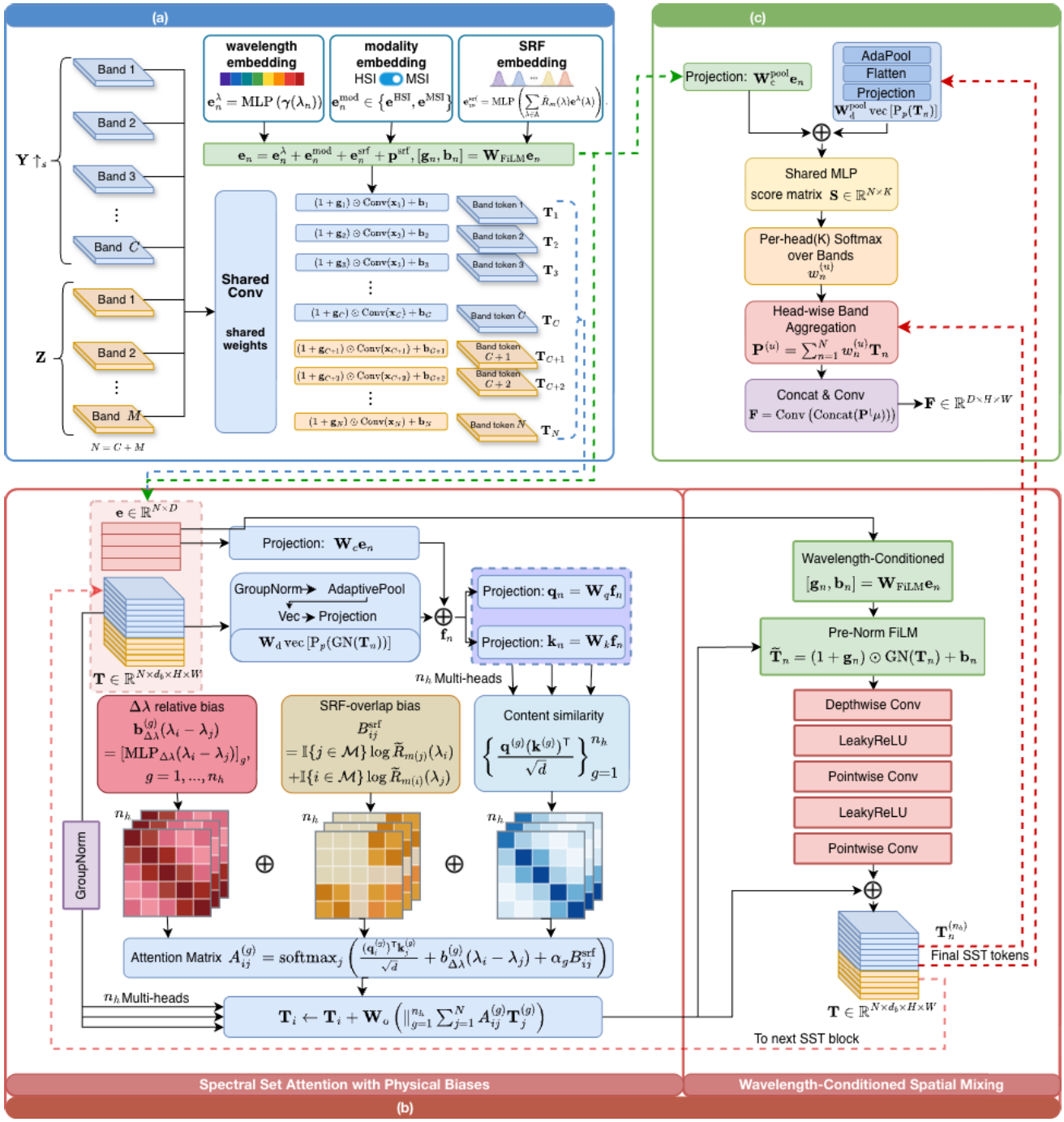


Fig. 4. Structure of the SST. (Top left) Band tokenization, where each HSI/MSI band is lifted by a shared convolution and modulated (FiLM) by its conditioning vector $\mathbf{e}_n$, which combines the wavelength embedding, a modality embedding, the SRF embedding of MSI bands, and a global sensor prompt (6). (Bottom) One SST block, alternating 1) spectral set attention along the band axis, whose logits are augmented by the continuous $\Delta\lambda$ relative bias and the SRF-overlap physical bias; and 2) wavelength-conditioned spatial mixing with shared depthwise–pointwise convolutions under FiLM modulation. (Top right) Multihead attention pooling aggregates the N band tokens into the fused feature map $\mathbf{F}$. (a) Wavelength-conditioned band tokenization. (b) SST blocks $\times n_b$. (c) Multihead attention pooling.

providing a relative spectral position encoding that, unlike index-based relative positions, remains meaningful across sensors, and $\alpha^{(g)}$ is a learnable per-head scale on the SRF-overlap bias

$$B^{\mathrm{srf}}_{ij} = \mathbb{I}\left[j \in \mathcal{M}\right] \log \tilde{R}_{m(j)}(\lambda_i) + \mathbb{I}\left[i \in \mathcal{M}\right] \log \tilde{R}_{m(i)}\left(\lambda_j\right) \quad (10)$$

with $\tilde{R}_m(\lambda) = \max(\hat{R}_m(\lambda)/\max_{\lambda'} \hat{R}_m(\lambda'), \epsilon)$ the max-normalized SRF, evaluated at the token wavelengths by linear interpolation and floored at $\epsilon = 10^{-3}$ before the logarithm, which keeps the bias and its gradients finite ($\geq \log \epsilon \approx -6.9$); $\hat{R}$ is the Stage-1 estimate during adaptation and the nominal SRF during pretraining, and the prior acts symmetrically on the HSI→MSI and MSI→HSI directions. This bias injects a routing prior: an HSI band at wavelength λ_i preferentially attends to the MSI bands whose spectral response actually covers λ_i, which is the physical pathway

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

through which high-resolution (HR) spatial detail should flow.

Unlike standard attention, the values are not projected vectors but the FR token feature maps themselves: the d_b channels are split into n_h head groups, and each head mixes its group across bands

$$\mathbf{T}_i \leftarrow \mathbf{T}_i + \mathbf{W}_{\mathrm{o}} \left(\Big\|_{g=1}^{n_h} \sum_{j=1}^{N} A_{ij}^{(g)} \, \mathbf{T}_j^{(g)} \right) \tag{11}$$

where $\mathbf{T}_j^{(g)} \in \mathbb{R}^{(d_b/n_h)\times H\times W}$ is the gth channel group of the group-normalized token j, $\|$ denotes channel concatenation, and $\mathbf{W}_{\mathrm{o}}$ is a shared output projection along the channel dimension. Every band thus receives an FR mixture of all other bands weighted by physically informed affinities. The query–value split of (8) and (11) also fixes the cost profile of the block: computing the logits costs $\mathcal{O}(N^2 d)$ independently of the spatial size, whereas the value mixing costs $\mathcal{O}(N^2 d_b HW)$ and dominates the memory footprint, motivating the random-patch training and tiled full-image inference protocol of Section IV. The parameter count is independent of N: the complete fusion network has 3.50-M parameters in the production configuration (backbone 2.44 M, wavelength-query decoder 0.86 M, SRF embedding 0.20 M), and the two Stage-1 degradation MLPs of Section III-B add about 28k.

3) Wavelength-Conditioned Spatial Mixing: Spectral attention is complemented by a spatial mixing sublayer: the group-normalized token is FiLM-modulated by $\mathbf{e}_n$ and passed through a depthwise–pointwise convolution pair whose weights are shared across all bands

$$\mathbf{T}_n \leftarrow \mathbf{T}_n + \mathrm{PW}\left(\mathrm{DW}\left((1+\mathbf{g}_n)\odot \mathrm{GN}\left(\mathbf{T}_n\right) + \mathbf{b}_n\right)\right) \tag{12}$$

where the FiLM parameters $[\mathbf{g}_n, \mathbf{b}_n]$ are generated from $\mathbf{e}_n$ by a separate head as in (7), DW is a 3×3 depthwise convolution, and PW is a two-layer pointwise block with a $2\times$ channel expansion (LeakyReLU activations after DW and between the pointwise layers). Sharing the spatial filters keeps the operator band-agnostic, while the wavelength-dependent modulation lets different spectral regions develop different spatial behavior. Each SST block stacks one spectral attention and one spatial mixing sublayer [both prenormalized residual updates, cf. (8) and (12)], and the backbone stacks n_b blocks.

4) Multihead Attention Pooling: After the last block, the N tokens must be aggregated into a fixed-size fused feature $\mathbf{F} \in \mathbb{R}^{D\times H\times W}$. A single softmax pooling over bands would collapse the spectrum into one convex combination, a rank-one bottleneck that discards most spectral diversity. We instead score every token with K independent heads and concatenate the head-wise aggregates

$$\mathbf{F} = \mathrm{Conv}\left(\Big\|_{u=1}^{K} \sum_{n=1}^{N} w_n^{(u)} \, \mathbf{T}_n \right) \tag{13}$$

where the head-u weights $w_n^{(u)} = *softmax_n \psi_u(\mathbf{T}_n, \mathbf{e}_n)$ are produced by a shared scoring MLP ψ with K outputs, applied to the descriptor–condition fusion $\mathbf{W}_{\mathrm{d}}^{\mathrm{pool}} \mathrm{vec}(\mathrm{P}_p(\mathbf{T}_n)) + \mathbf{W}_{\mathrm{c}}^{\mathrm{pool}} \mathbf{e}_n$ constructed as in (8), and $\|$ denotes channel concatenation. The ablation in Section IV-F confirms the benefit of multihead over single-head pooling.

D. Wavelength-Query Decoding and Spectral Residual

1) Wavelength-Query Decoder: Output bands are generated by querying the fused feature with target wavelengths. For each target wavelength λ_c^{out}, its embedding (5) is first enriched by a single-head cross-attention that queries the MSI SRF embeddings of (6), $\tilde{\mathbf{e}}_c = \mathbf{e}^{\lambda}(\lambda_c^{\mathrm{out}}) + \mathrm{CA}(\mathbf{e}^{\lambda}(\lambda_c^{\mathrm{out}}), \{\tilde{\mathbf{e}}_m^{\mathrm{srf}}\}_{m=1}^{M})$, informing each query of the broadband observations that actually cover it; the enriched embedding is then broadcast-added to $\mathbf{F}$ and decoded by a shared convolutional head

$$\hat{\mathbf{X}}_c^{\mathrm{net}} = \mathrm{Conv}_{\mathrm{dec}}\left(\mathbf{F} + \tilde{\mathbf{e}}_c\right). \tag{14}$$

Since the queries are arbitrary wavelengths and the spatial size of $\mathbf{F}$ follows the HrMSI grid (the LrHSI is upsampled by bicubic interpolation at the input, accommodating any fractional s), the decoder synthesizes any number of output bands at any spatial scale without architectural change. Queries are processed in chunks to bound memory.

2) Confidence-Masked Spectral Residual: The network predicts a residual on top of a physically motivated baseline: the bicubic-upsampled LrHSI linearly interpolated along wavelength to the target wavelengths, denoted $\mathcal{I}_\lambda(\mathbf{Y}\uparrow_s)$. This baseline is reliable only where a target wavelength lies close to an observed one, so it is gated per band by a confidence mask

$$\hat{\mathbf{X}}_c = \hat{\mathbf{X}}_c^{\mathrm{net}} + \rho_c \, \left[\mathcal{I}_\lambda\left(\mathbf{Y}\uparrow_s\right)\right]_c, \quad \rho_c = \sigma\left(\kappa \, \tfrac{g_{\max} - d_c}{g_{\max}}\right) \tag{15}$$

where d_c is the distance from λ_c^{out} to the nearest input wavelength, $g_{\max}$ is the trust radius, and $\sigma(\cdot)$ is the sigmoid function. The mask makes the wavelength-coverage boundary explicit: targets far from all observed wavelengths receive no interpolated prior and must be inferred purely by the network; if no spectral overlap exists at all, recovery is impossible, as quantified in Section IV-D.

E. Training Paradigm

1) FR Consistency: With the degradation frozen, the network output at full resolution is constrained to reproduce both observations

$$\begin{aligned}\mathcal{L}_{\mathrm{FR}} = {} & \left\| \left(\hat{\mathbf{X}} * k_{\hat{\theta}}\right)\downarrow_s - \mathbf{Y} \right\|_1 + \lambda_{\mathrm{spec}} \left\| \hat{\mathbf{R}}\hat{\mathbf{X}} - \mathbf{Z} \right\|_1 \\ & + \lambda_{\mathrm{grad}} \, \mathcal{L}_{\mathrm{grad}}\left(\hat{\mathbf{X}}, \mathbf{Z}\right) + \lambda_{\mathrm{tv}} \, \mathrm{TV}\left(\hat{\mathbf{X}}\right)\end{aligned} \tag{16}$$

where $\mathcal{L}_{\mathrm{grad}}$ aligns the spatial gradients of the band-averaged prediction with those of the band-averaged HrMSI, and TV$(\cdot)$ is an anisotropic total-variation penalty. No HrHSI enters (16).

2) Merged RR Supervision: Consistency losses alone constrain $\hat{\mathbf{X}}$ only through degraded projections. Following the Wald protocol logic, the learned operators can also synthesize a *supervised* problem one scale down: degrading the observations to $\tilde{\mathbf{Y}} = (\mathbf{Y} * k_{\hat{\theta}})\downarrow_s$ and $\tilde{\mathbf{Z}} = (\mathbf{Z} * k_{\hat{\theta}})\downarrow_s$ makes $\mathbf{Y}$ itself the ground truth for the RR prediction $\hat{\mathbf{X}}^{\mathrm{rr}} = f_\Theta(\tilde{\mathbf{Y}}, \tilde{\mathbf{Z}})$

$$\mathcal{L}_{\mathrm{RR}} = \left\| \hat{\mathbf{X}}^{\mathrm{rr}} - \mathbf{Y} \right\|_1 + \lambda_{\mathrm{spec}}^{\mathrm{rr}} \left\| \hat{\mathbf{R}}\hat{\mathbf{X}}^{\mathrm{rr}} - \tilde{\mathbf{Z}} \right\|_1. \tag{17}$$

Rather than optimizing the two objectives sequentially, which induces an abrupt regime shift when switching stages, we *merge* them: every iteration performs one RR forward pass and one FR forward pass and minimizes

$$\mathcal{L} = \mathcal{L}_{\mathrm{FR}} + \lambda_{\mathrm{rr}} \, \mathcal{L}_{\mathrm{RR}}. \tag{18}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

The RR branch anchors the network with direct (self-generated) supervision while the FR branch optimizes the actual target domain; their joint optimization improves accuracy by + 1.04 dB over the sequential schedule at only 10% extra time (see Section IV-F).

3) Universal Pretraining and Per-Scene Adaptation: Because the backbone is band-agnostic, one set of weights can be optimized across sensors. In the pretraining stage, each step draws one source dataset uniformly at random from a heterogeneous pool (equal probability for every dataset, independent of its image size), crops random patches, synthesizes the observation pair, and minimizes the same consistency objective (16), evaluated with the nominal synthesis operators of the source data (the Gaussian PSF and S2B SRF with which its observations are simulated). The reference HrHSI is used only to synthesize the inputs and never enters the loss gradients, so pretraining sees the same observation pairs as any per-image unsupervised method applied to the pool scenes, together with their nominal degradation metadata. We deliberately use these nominal operators: they are sensor metadata rather than HrHSI labels, they give every pool scene a fixed consistency target throughout pretraining, and they never reach the target scene, whose degradation Stage 1 estimates from its own observations; pretraining with per-source learned operators is left for future work. To fuse a new scene, AnyFusion runs Stage 1 on the target observations, loads the pretrained backbone, and performs the merged Stage 2 adaptation, which, warm-started by the universal representation, matches the final quality of FS optimization with 2.0–3.3× fewer iterations (see Section IV-C). Throughout this article, "universal" refers to sensor-configuration universality (independence from band count, wavelength layout, SRF, and scale) rather than to coverage of all scene distributions; enlarging the pretraining pool is an orthogonal direction discussed in Section V.

IV. Experiments

A. Datasets and Experimental Setup

1) Datasets: We evaluate the proposed AnyFusion framework on four hyperspectral benchmark datasets covering diverse sensor characteristics and spectral ranges. Table I summarizes the key properties of each dataset.

The **Pavia University (PaviaU)** dataset [76] was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS-3) airborne sensor over Pavia, Italy, containing 103 spectral bands after removing noisy channels. The **Houston18** dataset [77] was collected by the ITRES Compact Airborne Spectrographic Imager (CASI) for the 2018 IEEE GRSS Data Fusion Contest, with 46 bands spanning the visible to near-infrared (NIR) range. The **Chikusei** dataset [78] was captured by a Headwall Hyperspec-VNIR-C sensor over Chikusei, Japan, covering 128 bands. The **Tiangong (TG)** dataset [79] was acquired by the hyperspectral imager onboard the TG-1 space laboratory, containing 54 bands in the visible to NIR range. For each dataset, a square subimage of the size listed in Table I is cropped from the original scene and serves as the ground-truth HrHSI.

2) Simulation Protocol: Following the standard Wald protocol, the original HSI serves as the ground-truth HrHSI. The LrHSI is generated by applying a Gaussian PSF with $\sigma = s/2.35482$ HR pixels (i.e., a full width at half maximum of one LR pixel) followed by spatial downsampling by the scale factor s ($s = 4$ unless stated otherwise). The HrMSI is synthesized by convolving the HrHSI with the S2B SRF, producing 8–10 multispectral bands depending on the HSI wavelength coverage. During training, the PSF and SRF are treated as unknown: Stage 1 learns the degradation operators from the observed LrHSI and HrMSI in a fully unsupervised manner.

3) Training Configuration: The proposed method uses a two-stage training pipeline. Stage 1 learns the PSF and SRF degradation operators (2000 iterations, on a central patch of the training patch size for the simulated benchmarks and on the full image for the real pair of Section IV-E). Stage 2 jointly optimizes the fusion network in a merged RR and FR mode for 4000 iterations. The SST backbone uses $d_b = 16$ band dimension, four fusion blocks, and hidden dimension $D = 256$. Training employs random patch cropping (PaviaU: 256 × 256; Houston18, TG, and Chikusei: 192 × 192) with cosine learning rate scheduling and automatic mixed precision (AMP). Evaluation is performed on the full image via tiled inference with cosine-windowed overlap blending. All experiments are conducted on a single NVIDIA RTX 3090 GPU (24 GB).

All stages are optimized with Adam. Stage 1 uses a learning rate of 10^{-3} with regularization weights $\eta_k = 10^{-4}$ and $\eta_r = 10^{-3}$ in (4); the PSF prior width is $\sigma_0 = 0.425$ LR pixels and the SRF prior bandwidth is $\beta = 80$ nm. Stage 2 uses a learning rate of 5×10^{-5} with loss weights $\lambda_{\text{spec}} = 8$, $\lambda_{\text{grad}} = 1$, $\lambda_{\text{tv}} = 10^{-4}$, $\lambda_{\text{rr}} = 1$, and $\lambda_{\text{spec}}^{\text{rr}} = 2$; the spectral weight was selected by a short-budget screening over $\lambda_{\text{spec}} \in \{4, 8, 16\}$ on PaviaU and validated end-to-end at the full budget. All hyperparameters are fixed once across all datasets and experiments; the only dataset-specific settings are the training patch size (bounded by image size and GPU memory) and the reduced TG adaptation rate noted in Section IV-C, which, unlike all other settings, was selected with the reference (see the footnote of the table therein). No per-scene tuning is performed at deployment. The spectral residual (15) uses a trust radius $g_{\max} = 20$ nm with $\kappa = 5$. The spectral set attention uses $n_h = 4$ heads with a total query–key dimension of 128 (i.e., $d = 32$ per head) and a descriptor pooling window of $p = 8$; the pooling stage uses $K = 8$ heads. The two degradation networks of Stage 1 are lightweight four-layer MLPs (hidden width 64, SiLU activations, and six Fourier frequencies) with zero-initialized output layers, so optimization starts exactly from the Gaussian priors. Tiled full-image inference uses tiles of the training patch size with a 32-pixel cosine-ramp overlap. Universal pretraining runs 10^4 steps with a learning rate of 10^{-4} under the same cosine schedule, sampling 144 × 144 patches with a base batch size of 4 patches per step (2 for the Houston18 leave-one-out (LOO) checkpoint), halved for datasets with more than 60 bands to fit the 24-GB GPU. The four LOO checkpoints are pretrained without the spectral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS ON FOUR BENCHMARK DATASETS (S2B SRF, SCALE = 4, FULL-IMAGE EVALUATION). ALL ROWS REPORT MEAN±STD OVER THREE INDEPENDENT RUNS UNDER THE IDENTICAL PROTOCOL: OUR ROWS OVER RANDOM SEEDS 42/7/123, AND EACH BASELINE OVER ITS ORIGINAL-SEED RUN PLUS TWO RESEEDED RUNS (SEEDS 7/123); CUCANET (†) USES SEEDS 42/7/123 (SEE FOOTNOTE). THE BEST MEAN RESULT IN EACH COLUMN IS IN BOLD, AND THE SECOND BEST IS UNDERLINED; TIES AT THE DISPLAYED PRECISION ARE RESOLVED, AND ALL MARGINS QUOTED IN THE TEXT COMPUTED, AT FULL PRECISION BEFORE ROUNDING. ALL METHODS ARE EVALUATED UNDER THE UNSUPERVISED SETTING WITHOUT ACCESS TO GROUND-TRUTH HRHSI DURING TRAINING; ALL BUT CNMF (GIVEN THE NOMINAL SRF, SECTION IV-A) ALSO ESTIMATE THE DEGRADATION BLINDLY

		PaviaU			Houston18			Chikusei			TG		
Type	**Method**	PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓
Trad.	CNMF	34.34±0.30	4.93±0.14	3.32±0.12	41.57±0.18	2.85±0.03	1.53±0.02	43.30±0.48	2.73±0.09	4.89±0.46	36.53±0.26	2.69±0.06	1.54±0.05
Unsup. DL	CUCaNet†	37.04±0.17	2.98±0.05	2.50±0.10	41.98±0.18	2.25±0.04	2.10±0.02	44.93±0.27	2.10±0.11	3.26±0.24	38.61±1.40	1.42±0.02	1.27±0.19
	FusionNet	33.64±1.51	4.91±0.45	3.51±0.44	33.40±0.12	3.52±0.10	3.94±0.06	40.13±0.13	3.28±0.12	4.61±0.05	30.72±0.55	2.71±0.08	2.97±0.19
	UDALN	34.65±0.06	4.30±0.13	3.15±0.04	34.23±0.92	3.38±0.13	3.54±0.49	42.06±0.30	2.43±0.18	4.01±0.14	32.77±0.45	2.25±0.19	2.36±0.11
	DTDNML	35.69±0.28	4.05±0.07	2.93±0.06	35.18±0.17	3.10±0.12	3.40±0.17	43.87±0.45	2.40±0.04	3.78±0.31	34.33±0.42	1.73±0.06	2.11±0.06
	MIAE	39.65±0.92	2.95±0.16	2.52±0.85	44.53±0.12	2.57±0.10	2.15±0.18	**48.81**±0.37	2.85±0.27	7.66±1.72	43.07±0.67	1.31±0.11	1.09±0.34
	CS2DIPs	39.62±0.16	**2.78**±0.07	1.97±0.09	40.69±0.01	3.10±0.07	2.04±0.05	45.58±0.34	1.79±0.06	3.47±0.05	39.32±0.36	1.82±0.09	1.14±0.07
	EDIP-Net	34.80±0.16	4.09±0.07	2.90±0.16	40.12±0.24	2.54±0.13	1.86±0.11	46.11±1.00	1.66±0.14	**2.36**±0.06	37.39±1.00	2.09±0.07	1.69±0.08
Ours	from-scratch	40.10±0.36	3.01±0.06	1.93±0.11	44.73±0.88	1.79±0.07	1.25±0.09	45.84±1.72	1.81±0.24	3.77±0.56	43.06±0.92	1.28±0.08	0.87±0.11
	pretrain	**40.72**±0.05	3.05±0.03	**1.90**±0.02	**45.75**±0.03	**1.67**±0.00	**1.07**±0.01	48.61±0.44	**1.54**±0.05	2.99±0.06	**43.73**±0.08	**1.18**±0.02	**0.74**±0.01

Our models use random-patch training with tiled full-image evaluation (FR = 4000); CS2DIPs and EDIP-Net run their official full-image optimization pipelines on identical observations. †CUCaNet's original small-magnitude random endmember initialization degenerates on TG to a per-band-constant solution (23.44 dB, bit-identical across seeds); we therefore initialize its endmembers from observed pixel spectra, in the manner of vertex component analysis, applied uniformly to all four datasets, which improves it on every dataset, and report three re-seeded runs of the fixed configuration.

residual of (15), which is enabled at adaptation; the all-dataset checkpoint of Section IV-E includes it.

4) Evaluation Metrics: Three reference-based metrics are reported: peak signal-to-noise ratio (PSNR, dB, higher is better), spectral angle mapper (SAM, degrees, lower is better), and relative dimensionless global error in synthesis (ERGAS, lower is better). PSNR is computed as the per-band average with max_val = 1.0. For the real-data experiment without a reference (see Section IV-E), these metrics are evaluated in an observation-consistency form, complemented by the structural similarity index (SSIM) [80] and the quality with no reference (QNR) index, as defined there.

5) Compared Methods: We compare AnyFusion against eight representative HSI–MSI fusion methods: one traditional matrix factorization method, CNMF [16], and seven unsupervised deep learning methods: CUCaNet [34], FusionNet [47], UDALN [48], MIAE [36], DTDNML [50], CS2DIPs [51], and EDIP-Net [42]. CS2DIPs couples spatial and spectral DIPs through a nonnegative matrix-vector tensor factorization, and EDIP-Net is a four-stage enhanced DIP pipeline with blind degradation estimation; both run with their official implementations. All methods use the same S2B SRF and 4× spatial downsampling protocol and observe identical LrHSI/HrMSI inputs; all learning-based methods estimate the PSF and SRF blindly, whereas CNMF, which has no degradation-estimation step in our implementation, is given the nominal SRF (an advantage to it) and fuses the bicubic-upsampled LrHSI without a PSF model. All methods run with a single fixed configuration (official defaults where available) across all datasets, with no per-dataset tuning. A direct comparison with the supervised band-agnostic framework SSA [43] is not possible: it requires HrHSI training targets, and its code is not publicly available at the time of writing; the channel-index ablation of Section IV-D4 therefore reimplements its ordinal band-addressing semantics within our own framework as the closest controlled proxy.

B. Comparison With State-of-the-Art Methods

1) Quantitative Results: Table II presents the quantitative comparison across all four benchmark datasets. Our method is evaluated under two settings: *FS* (per-image unsupervised optimization without pretraining) and *pretrain* (PT; universal pretraining followed by per-image unsupervised adaptation); the abbreviations FS and PT are used throughout the remaining tables and figures.

Several observations can be drawn from Table II.

a) Overall performance: Our pretrained model achieves the highest mean PSNR on three of the four benchmarks. On Houston18 the margin is largest: + 1.22 dB over the strongest baseline MIAE (45.75 ± 0.03 versus 44.53 ± 0.12 dB), ranking first on all three metrics with our FS model second on all three. On PaviaU the margin is + 1.08 dB (40.72 ± 0.05 versus 39.65 ± 0.92 dB), exceeding MIAE's own run-to-run std. On TG the mean margin is + 0.65 dB (43.73 ± 0.08 versus 43.07 ± 0.67 dB, with a TG-specific adaptation rate; see the footnote of Table III); this lies within MIAE's run-to-run variance, but our method also wins SAM and ERGAS outright (1.18 versus 1.31°; 0.74 versus 1.09). The pretrained rows are moreover highly reproducible, with a PSNR seed std of at most 0.08 dB on PaviaU, Houston18, and TG, and 0.44 dB on Chikusei (versus 1.72 dB from scratch).

b) Chikusei analysis: MIAE attains the highest mean PSNR on Chikusei (48.81 ± 0.37 versus our 48.61 ± 0.44 dB), a −0.20-dB gap within both methods' run-to-run std. Its spectral fidelity, however, is markedly worse and unstable across runs (SAM 2.85 ± 0.27°; ERGAS 7.66 ± 1.72), whereas our method attains the best SAM (1.54°) and second-best ERGAS (2.99) in the table. MIAE's low-rank

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

TABLE III

PRETRAIN ADAPTATION EFFECTIVENESS (ALL QUANTITIES TILED FULL-IMAGE). "FINAL PSNR" IS THE THREE-SEED MEAN±STD FROM TABLE II. "MATCH-FS ITER" IS THE ITERATION AT WHICH THE PRETRAINED ADAPTATION FIRST REACHES THE FINAL (4000-ITERATION) PSNR OF THE FS RUN, READ FROM THE SEED-42 CONVERGENCE CURVES OF FIG. 11; "SPEEDUP" IS 4000 DIVIDED BY THIS ITERATION. "INIT GAIN" IS THE PT–FS PSNR DIFFERENCE AT THE FIRST EVALUATION (ITERATION 200). Δ IS COMPUTED AT FULL PRECISION BEFORE ROUNDING

	Final PSNR (dB)					
Dataset	FS	PT	Δ	**Match-FS iter**	**Speedup**	**Init gain**
PaviaU	40.10±0.36	**40.72**±0.05	+0.62	1200	3.3×	+9.7 dB
Houston18	44.73±0.88	**45.75**±0.03	+1.03	2000	2.0×	+6.4 dB
Chikusei	45.84±1.72	**48.61**±0.44	+2.77	1400	2.9×	+7.2 dB
TG	43.06±0.92	**43.73**±0.08	+0.66	2000	2.0×	+4.4 dB

TG uses a reduced adaptation rate (2×10^{-5}; from-scratch runs keep 5×10^{-5}). At the default rate the pretrained TG run peaks at 43.75 dB (iteration 1600) and then drifts to 43.34 dB with SAM rising to 1.42°, a pretraining gain of −0.10 dB (seed 42). The rate was selected once on TG from $\{1, 2, 5\} \times 10^{-5}$ using the reference, a deviation from the reference-free protocol; it is inferior on the other datasets and used nowhere else.

spectral modeling is well matched to this spectrally redundant 128-band dataset, but its advantage does not extend beyond PSNR.

c) Baseline comparison: Among the baselines, MIAE is overall the strongest competitor, holding the best baseline mean PSNR on all four datasets (on PaviaU essentially tied with CS2DIPs), while FusionNet and UDALN show limited performance across all datasets (e.g., FusionNet: 33.64 dB on PaviaU, 33.40 dB on Houston18). DTDNML, despite using deep Tucker decomposition, achieves moderate results (34–44 dB). CUCaNet is the second-strongest baseline in PSNR on Houston18 (41.98 dB) once its endmember initialization is stabilized (see Table II, footnote); its sensitivity to initialization scale (the original scheme degenerates to a constant solution on TG) nonetheless illustrates how fragile per-image unmixing optimization can be under unfamiliar sensor configurations. The two most recent DIP-based baselines behave complementarily: CS2DIPs is strong on PaviaU (39.62 dB, with the best SAM of 2.78°) but drops well below MIAE and our method on Houston18 (−5.06 dB versus ours) and TG (−4.41 dB); EDIP-Net attains the best ERGAS on Chikusei (2.36) yet is not competitive on PaviaU (34.80 dB). Neither recent method ranks first in PSNR on any dataset, confirming that per-image DIP optimization alone does not close the gap to model-inspired (MIAE) or pretrained band-agnostic (ours) designs.

d) FS competitiveness and variance: Even without pretraining, our FS model already matches or exceeds the strongest baseline on three datasets: it slightly surpasses MIAE on PaviaU (40.10 ± 0.36 versus 39.65 ± 0.92 dB), exceeds it on Houston18 (44.73 ± 0.88 dB, above the best baseline mean of 44.53 ± 0.12 dB), and essentially matches it on TG (43.06 ± 0.92 versus 43.07 ± 0.67 dB). Its seed variance, however, is substantial (up to ±1.72 dB on Chikusei), reflecting the optimization instability inherent to purely per-image unsupervised training, which pretraining removes [see (5)].

e) Consistent pretrain gains: Universal pretraining improves the mean PSNR on all four datasets while shrinking the seed std by 4–26×, so that per-image unsupervised fusion becomes both more accurate and more reproducible; the latter is important in deployment, where no reference is available to identify a poor run. Section IV-C quantifies these gains.

f) PSNR–SAM tradeoff on PaviaU: Pretraining raises PSNR on PaviaU but leaves the mean SAM slightly higher (3.05° versus 3.01°), and CS2DIPs holds the best SAM (2.78°). The shift is within seed variance (SAM decreases on two of the three seeds). Our residual gap to CS2DIPs lies in the visible range (5.69° versus 4.96° for its reference run) and in the darkest pixels, where small absolute errors produce large angles, whereas in the NIR range our SAM is lower (1.51° versus 1.60°); we therefore do not claim spectral-angle superiority on PaviaU.

2) Visual Comparison: Figs. 5–8 present the visual comparison across all compared methods on all four datasets. Each figure shows a false-color composite (top row; band wavelengths in the captions), the per-pixel SAM error map (middle row), and the per-pixel RMSE map (bottom row). Consistent with the quantitative results, our method produces spatially uniform SAM error maps on all four datasets and the lowest overall RMSE on PaviaU and Houston18.

On PaviaU (see Fig. 5), CNMF and UDALN exhibit visible spectral distortion along building edges, while FusionNet shows the highest overall error due to its limited model capacity. DTDNML improves over FusionNet but still lags behind MIAE. Our pretrained model yields visibly lower RMSE than MIAE in heterogeneous regions. On Houston18 (see Fig. 6), the advantage of our method is particularly clear in high-contrast boundary areas, where nearly all baselines produce a visible error halo around the bright building. On TG (see Fig. 7), CNMF and FusionNet exhibit elevated errors across the whole scene; compared with the strongest baseline MIAE, our error maps are visibly more uniform, consistent with the SAM and ERGAS margins in Table II. On Chikusei (see Fig. 8), MIAE leads in mean PSNR, but its SAM map exhibits extensive field-wise elevated errors across large portions of the scene, matching its worse spectral metrics in Table II.

3) Spectral Curve Comparison: Fig. 9 compares reconstructed spectral signatures at the fixed pixel locations listed in its caption (our curves from the seed-42 runs, CNMF and MIAE from their reference runs). The proposed method (both FS and pretrain) closely tracks the ground-truth spectrum across the full wavelength range and preserves the characteristic absorption and reflection features.

4) Per-Band PSNR Analysis: Fig. 10 shows the per-band PSNR across the full wavelength range for each dataset, revealing the spectral profile of reconstruction quality.

The per-band curves come from each method's single reference run. On PaviaU, our pretrained model exceeds MIAE on roughly two thirds of the bands, most clearly in the NIR (700–838 nm). On Houston18 the curves alternate, but our worst-band PSNR is 37.1 dB versus MIAE's 29.4 dB. On Chikusei, MIAE's curve dips to 25.8 dB near 714 nm, whereas ours stays above 41.4 dB, consistent with its better SAM. On

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

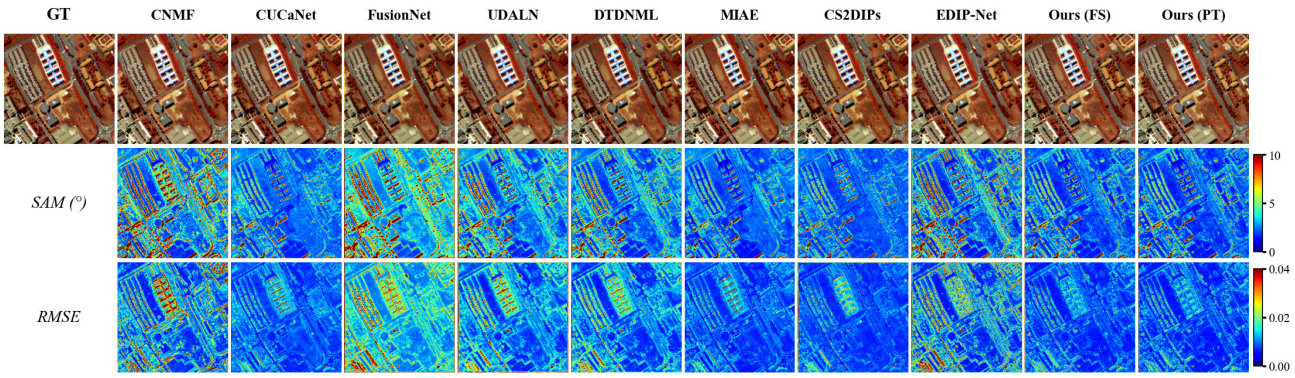


Fig. 5. Visual comparison on PaviaU. (Left to right) Ground truth, CNMF, CUCaNet, FusionNet, UDALN, DTDNML, MIAE, CS2DIPs, EDIP-Net, Ours (FS), Ours (pretrain). (Top) False-color composite (R/G/B = 718/594/490 nm; 2%–98% percentile stretch per panel); (middle) SAM error map; and (bottom) root-mean-square error (RMSE) map. All error maps within a row share the colorbar shown on the right.

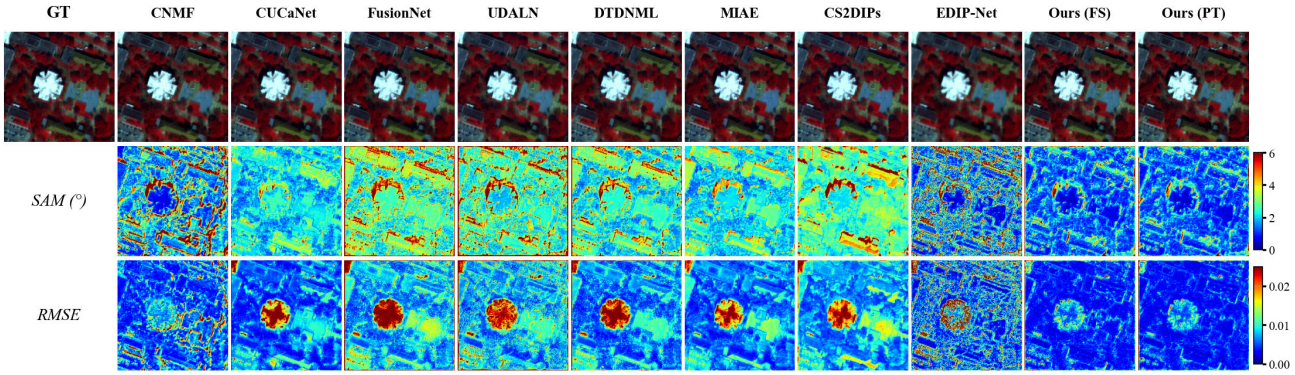


Fig. 6. Visual comparison on Houston18 (false-color R/G/B = 847/646/475 nm). Layout identical to Fig. 5. Our method achieves the lowest SAM and RMSE errors, particularly in high-contrast boundary areas.

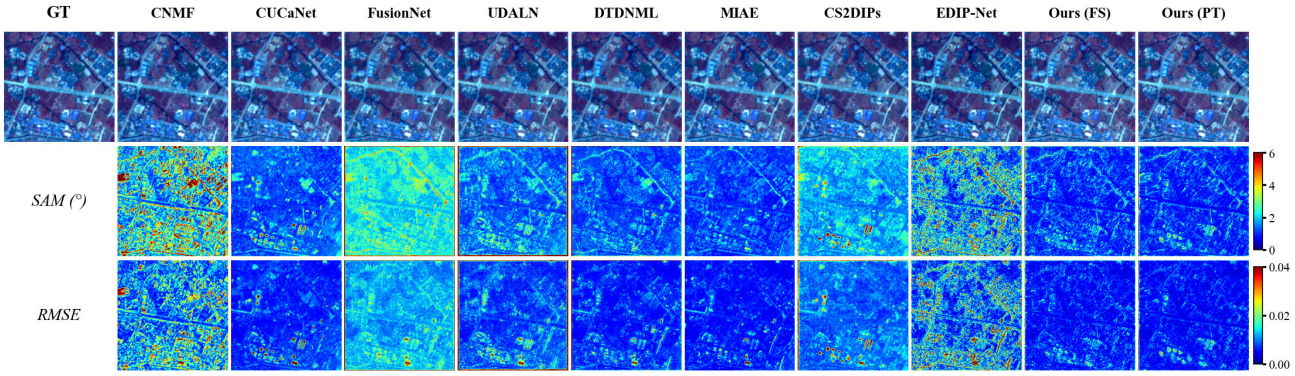


Fig. 7. Visual comparison on TG (false-color R/G/B = 721/578/479 nm). Layout identical to Fig. 5. CNMF and FusionNet show elevated errors across the scene, while our error maps are the most uniform.

TG the profiles are similar, with our advantages toward both ends of the spectrum.

C. Universal Pretraining Effectiveness

The main comparison already showed that the pretrained model consistently outperforms its FS counterpart; this section isolates and quantifies the benefits of universal pretraining. Pretraining follows an LOO protocol: when adapting to a target dataset, the checkpoint is pretrained on the other three datasets only, so the target sensor is strictly unseen during pretraining.

1) Quantitative Improvement: Table III reports the PSNR improvement and convergence speedup achieved by pretraining.

The results reveal three complementary benefits of universal pretraining as follows.

a) Consistent quality improvement: Pretraining improves the mean final full-image PSNR on all four datasets, with gains ranging from +0.62 dB (PaviaU) to +2.77 dB (Chikusei), while reducing the seed std by 4–26×. Unlike prior pretrain-then-finetune approaches that require HrHSI

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

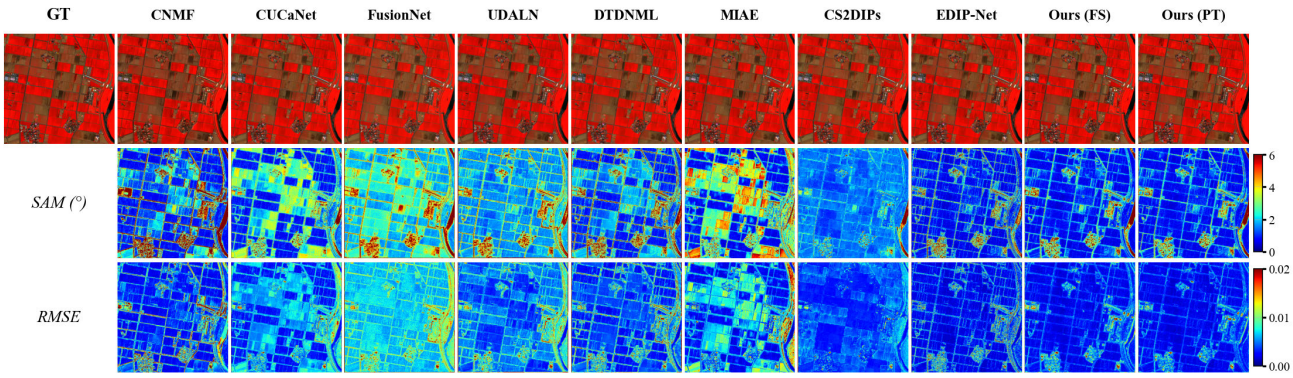


Fig. 8. Visual comparison on Chikusei (false-color R/G/B = 822/626/461 nm). Layout identical to Fig. 5. MIAE attains the highest mean PSNR on this dataset, while our method yields lower spectral-angle errors (see Table II).

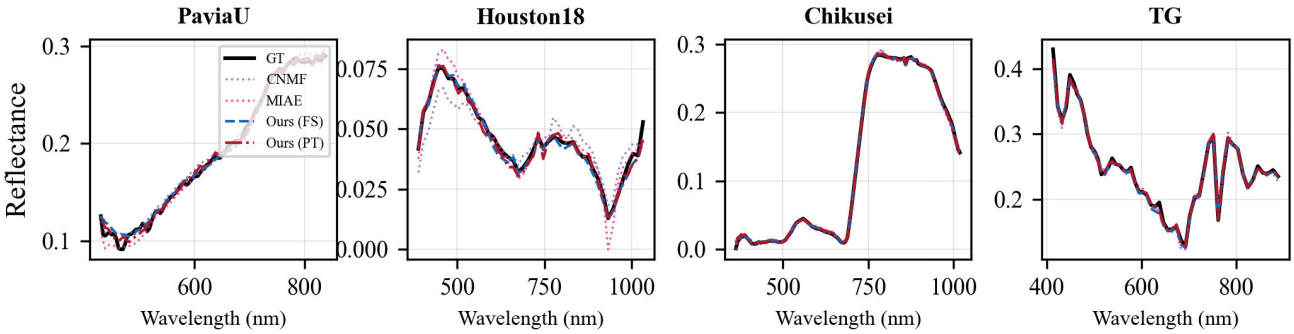


Fig. 9. Spectral curve comparison at fixed pixel locations, given as (row, column) in zero-based full-image coordinates: PaviaU (100, 100); Houston18, Chikusei, and TG (80, 80). Black solid: ground truth; dotted: baselines (CNMF, MIAE); blue dashed: Ours (FS); red dash-dot: Ours (pretrain). Our method closely tracks the ground-truth spectrum across the full wavelength range. (a) PaviaU, (b) Houston18, (c) Chikusei, and (d) TG.

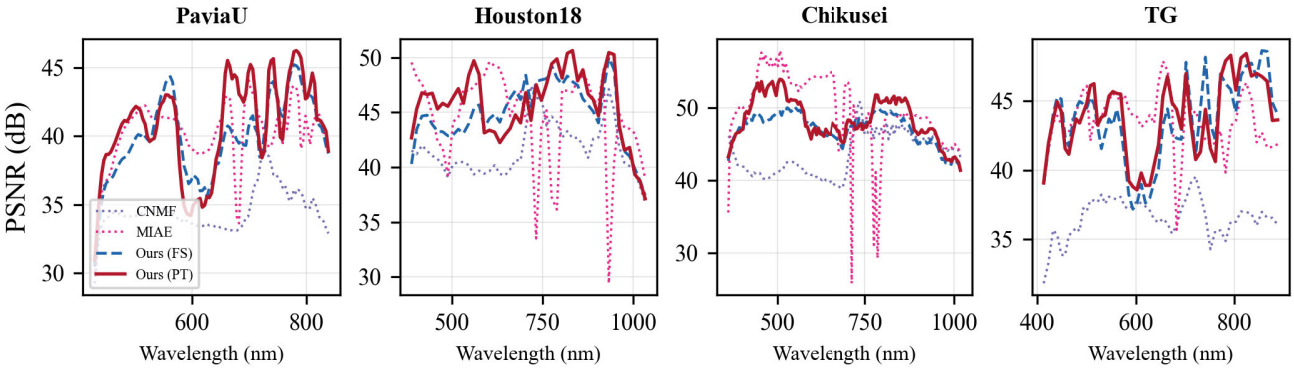


Fig. 10. Per-band PSNR comparison across the full wavelength range. (a) PaviaU (103 bands), (b) Houston18 (46 bands), (c) Chikusei (128 bands), and (d) TG (54 bands). Our method maintains a flatter per-band profile with a substantially higher worst-band PSNR than MIAE, most visibly on Houston18 and Chikusei.

labels, our gains are achieved through purely unsupervised adaptation.

b) Equal-quality speedup: Measured on the full-image convergence curves, the pretrained model reaches the *final* quality that FS training attains only after all 4000 iterations with 2.0–3.3× fewer iterations (1200 on PaviaU, 1400 on Chikusei, and 2000 on Houston18 and TG), and then continues to improve beyond it. Since the curves increase monotonically with diminishing returns, the residual gain can be traded for adaptation time along the pretrained curve; the wall-clock consequences are quantified in Section IV-H.

c) High initial quality: The pretrained model starts adaptation at a substantially higher full-image PSNR: +4.4 to +9.7 dB above FS at the first evaluation (iteration 200). On PaviaU, the pretrained model at iteration 200 (36.57 dB) already exceeds what the FS model reaches only between iterations 600 and 800, confirming that universal pretraining provides an effective spectral–spatial initialization that transfers across heterogeneous sensor configurations.

2) Convergence Curves: Fig. 11 shows the corresponding convergence trajectories: on all four datasets the pretrained

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

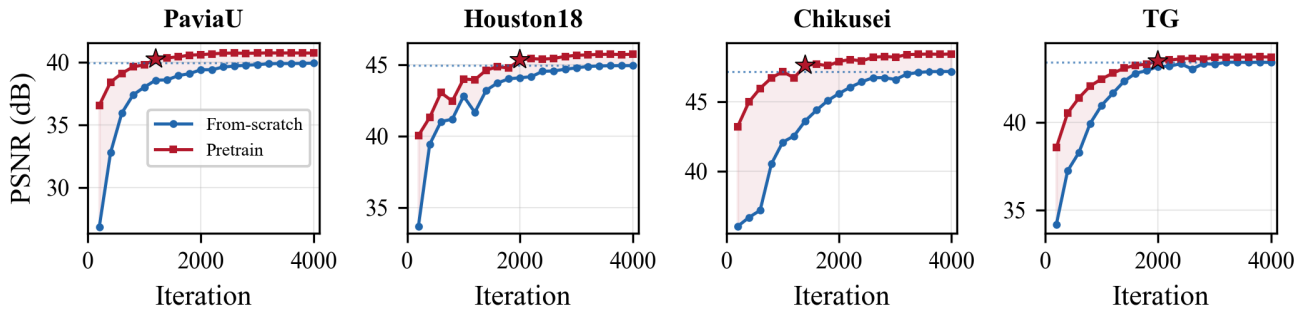


Fig. 11. PSNR convergence curves during unsupervised adaptation (tiled full-image evaluation every 200 iterations). Blue: FS; red: PT. The pretrained model starts at a substantially higher PSNR and converges faster on all four datasets. (a) PaviaU, (b) Houston18, (c) Chikusei, and (d) TG.

curve starts far above the FS one and remains above it throughout adaptation.

D. Cross-Configuration Generalization

Section IV-C established that the learned representation transfers across datasets; this section examines the complementary claim of "fuse anywhere": that a single architecture accommodates arbitrary sensor configurations. We evaluate four dimensions of configuration variability on PaviaU (spatial scale, input band subset, wavelength range, and SRF), each corresponding to a way in which a new sensor may differ from previously encountered ones. In all cases the network architecture is used without any modification. The cross-band and cross-wavelength experiments are run from scratch and thus demonstrate *architectural flexibility*; pretrained cross-sensor *transfer* is tested by the cross-scale and cross-SRF experiments, by the LOO protocol of Section IV-C, and by the real pair of Section IV-E.

1) Cross-Scale Generalization: Table IV (FS columns) reports fusion performance across seven spatial downsampling factors from 2× to 6×, including the noninteger factors 2.5× and 3.3×.

PSNR decreases overall from 41.42 dB (2×) to 38.53 dB (6×) as shown in Fig. 12, which is expected as larger downsampling factors discard more spatial information. The two sampling-grid families (filled versus hollow markers in Fig. 12) each follow a decreasing trend, with the fractional-grid points sitting slightly above the integer-grid curve; the decrease of ERGAS with scale in Table IV is largely an artifact of its $100/s$ prefactor. A single architecture handles all scale factors, integer and fractional alike, without any modification, owing to the arbitrary-scale input resampling and wavelength-query decoding of Section III-D.

Beyond architectural flexibility, deployment also requires that universally pretrained knowledge survive a change of spatial scale. To test this, the pretrained columns of Table IV adapt the *same* LOO checkpoint as Table II, pretrained exclusively at 4× on a pool that contains neither PaviaU nor any sensor with its band layout, at all seven adaptation scales from 2× to 6×, including the fractional factors, and compare against FS training under an identical protocol.

As summarized in Fig. 12(c), the pretraining gain is positive at *every* adaptation scale: it peaks at 3× (+1.81 dB), remains above +1.3 dB at 2× and 4×, and decays gracefully to +0.64 dB at 6×, the direction of increasing ill-posedness. The gain appears smallest at the fractional factors 2.5× and 3.3×, whose FS baselines are already elevated by the cleaner fractional-grid alignment; repeating both pairs with a second seed shows the FS results varying by 0.9–1.5 dB (with the per-seed paired gains rising to +2.17 and +1.37 dB) while the pretrained results vary by at most 0.11 dB. The small gains at these two factors therefore reflect favorable FS runs rather than weaker transfer. SAM improves in parallel at the integer scales 2×–4× and is essentially unchanged at 6× (3.60 versus 3.61°). Since the pretraining pool shares neither the scene, the sensor, nor the band layout with the adaptation task, every row constitutes a three-way configuration shift, compounded by a scale shift except at 4×, extending the "train once, fuse anywhere" claim to the scale dimension.

2) Cross-Band Generalization: Table V evaluates fusion performance with varying numbers of input HSI bands.

Uniform subsampling (every_3) causes only a modest 0.45-dB degradation, showing that the wavelength-aware encoder effectively interpolates between observed bands. A

TABLE IV

CROSS-SCALE GENERALIZATION AND PRETRAINING TRANSFER ON PAVIAU (S2B SRF; RANDOM-PATCH (192) TRAINING WITH TILED FULL-IMAGE EVALUATION). ALL CONFIGURATIONS USE THE SAME NETWORK ARCHITECTURE. THE PRETRAINED COLUMNS ADAPT THE 4× LOO CHECKPOINT OF TABLE II AT EVERY SCALE, WITH NO REPRETRAINING

Scale	From-scratch			Pretrain (4×)			ΔPSNR
	PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓	
2.0×	41.42	2.57	3.45	**42.78**	**2.35**	**3.00**	+1.36
2.5×‡	40.86	**2.79**	**2.87**	**41.42**	2.82	2.88	+0.56
3.0×	39.75	2.98	2.85	**41.56**	**2.71**	**2.26**	+1.81
3.3×‡	41.24	**2.74**	2.12	**41.57**	2.75	**2.10**	+0.33
4.0×	39.21	3.26	2.17	**40.57**	**3.00**	**1.88**	+1.36
5.0×‡	39.53	3.27	1.64	**40.33**	**3.21**	**1.62**	+0.80
6.0×	38.53	**3.60**	1.54	**39.17**	3.61	**1.52**	+0.64

‡Scales marked ‡ (2.5×, 3.3×, 5×) use the fractional-grid resampling path (5× because it does not divide the 192 training patch), whose resampling convention coincides with that of the operators of Section III-B, so that the upsampled LrHSI input carries no sampling-phase offset relative to the HrMSI (Section IV-F), which slightly favors these rows; their seed variance is discussed in the text. All cross-scale runs share a common 192 training patch (the largest size compatible with every scale factor), which is why the 4× from-scratch row differs from the 256-patch entry of Table III.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

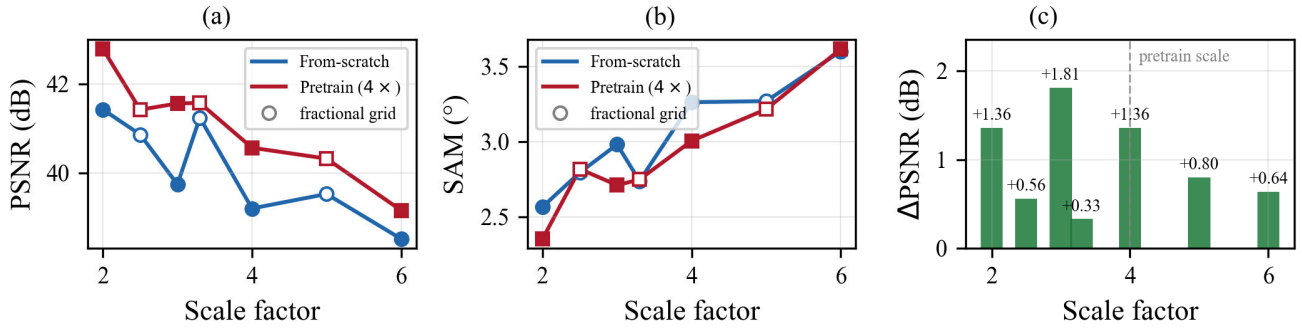


Fig. 12. Cross-scale generalization and pretraining transfer on PaviaU (S2B SRF, full-image evaluation). (a) PSNR and (b) SAM for FS versus 4×-pretrained adaptation across seven scale factors; hollow markers denote scales handled by the fractional-grid resampling path. (c) Pretraining gain ΔPSNR, positive at all seven scales and peaking at 3×.

TABLE V

CROSS-BAND GENERALIZATION ON PAVIAU (SCALE = 4, FS; RANDOM-PATCH (256) TRAINING WITH TILED FULL-IMAGE EVALUATION; "FULL" ROW IS THE SEED-42 REFERENCE RUN OF THE TABLE II PROTOCOL). "EVERY_3": EVERY 3RD BAND (35 BANDS); "RANDOM_40": A RANDOM 40% SUBSET (41 BANDS); "FIRST_HALF": BANDS 1–52 ONLY (430–634 NM). ALL CONFIGURATIONS RECONSTRUCT THE FULL 103-BAND OUTPUT

Config	Bands	PSNR↑	SAM↓	ERGAS↓
full	103	39.94	3.04	2.03
every_3	35	39.49	3.51	2.08
random_40	41	38.25	5.20	2.68
first_half	52	24.83	28.99	22.76

TABLE VI

CROSS-WAVELENGTH GENERALIZATION ON PAVIAU (SCALE = 4, FS; RANDOM-PATCH (256) TRAINING WITH TILED FULL-IMAGE EVALUATION; "FULL" ROW IS THE SEED-42 REFERENCE RUN OF THE TABLE II PROTOCOL). EACH HALF USES ONLY THE HSI AND MSI BANDS WITHIN THE SPECIFIED WAVELENGTH RANGE, AND BOTH THE RECONSTRUCTED OUTPUT AND THE METRICS ARE RESTRICTED TO THAT RANGE

Range	HSI bands	PSNR↑	SAM↓	ERGAS↓
Visible (430–634 nm)	52	39.01	5.75	2.63
NIR (638–838 nm)	51	42.97	1.60	0.90
Full (430–838 nm)	103	39.94	3.04	2.03

random 40% band subset degrades performance more substantially (−1.69 dB), as irregular spacing creates spectral gaps that are harder to interpolate.

The **first_half** configuration reveals a fundamental physical boundary: when the input LrHSI covers only the visible range (430–634 nm), the method cannot reconstruct NIR bands (638–838 nm) that have no spectral overlap with any input band. The resulting −15.11 dB collapse is a physical constraint rather than a model limitation: no information about the NIR domain exists in the input. The failure mode is also predictable in advance: the confidence mask of (15) identifies, from the wavelength metadata alone, which target bands lack spectral support (the visible-only experiment of Table VI below uses the same 52 input bands but reconstructs only the covered range).

Fig. 13 provides spatial visualizations. The first_half SAM map (rightmost column, middle row) shows catastrophic spectral distortion in the uncovered NIR region, while the full and every_3 configurations maintain consistently low error.

3) Cross-Wavelength Generalization: Table VI evaluates fusion on spectrally disjoint wavelength ranges: visible-only (52 bands, 430–634 nm, and three S2B MSI bands) and NIR-only (51 bands, 638–838 nm, and five S2B MSI bands). Unlike the cross-band experiment above, which always reconstructs the full 103-band output, each configuration here is a self-contained fusion task: the HSI and MSI inputs, the reconstructed output, and the evaluation are all restricted to the same wavelength range, so no extrapolation beyond the observed range is involved.

The NIR range achieves substantially higher PSNR (42.97 versus 39.01 dB) and much lower SAM (1.60° versus 5.75°) than the visible range. The gap stems from physical asymmetries between the two halves: 1) the NIR range has more MSI bands providing spatial guidance (5 versus 3) and 2) NIR spectra in this scene have inherently lower spectral variability. A single set of architectural hyperparameters accommodates both disjoint ranges and reflects these physical differences, asymmetries that a channel-index encoding [43] cannot express, since it carries no notion of which wavelengths are observed. This contrast is only indirect evidence for wavelength anchoring; Section IV-D4 provides the direct single-variable test.

4) Physical Wavelength Embedding Versus Channel Index: The cross-wavelength result of Table VI argues *indirectly* that anchoring spectral reasoning to physical wavelengths, rather than channel indices, is what enables cross-sensor transfer. We now provide a *direct*, single-variable test. Within our own framework we replace only the per-band conditioning embedding, swapping the physical-wavelength Fourier encoding of (5) for a learnable per-index (ordinal) embedding table, in which the kth band in the stack indexes the kth row regardless of its wavelength. This mirrors the ordinal addressing semantics of the Matryoshka kernel slicing in SSA [43], while every other component (the $\Delta\lambda$ relative bias, the SRF-overlap bias, and the entire backbone) is held fixed. We then run both the FS and the pretrain-then-adapt (PT) pipelines identically to the main experiments, over the same three seeds; Table VII reports the results.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

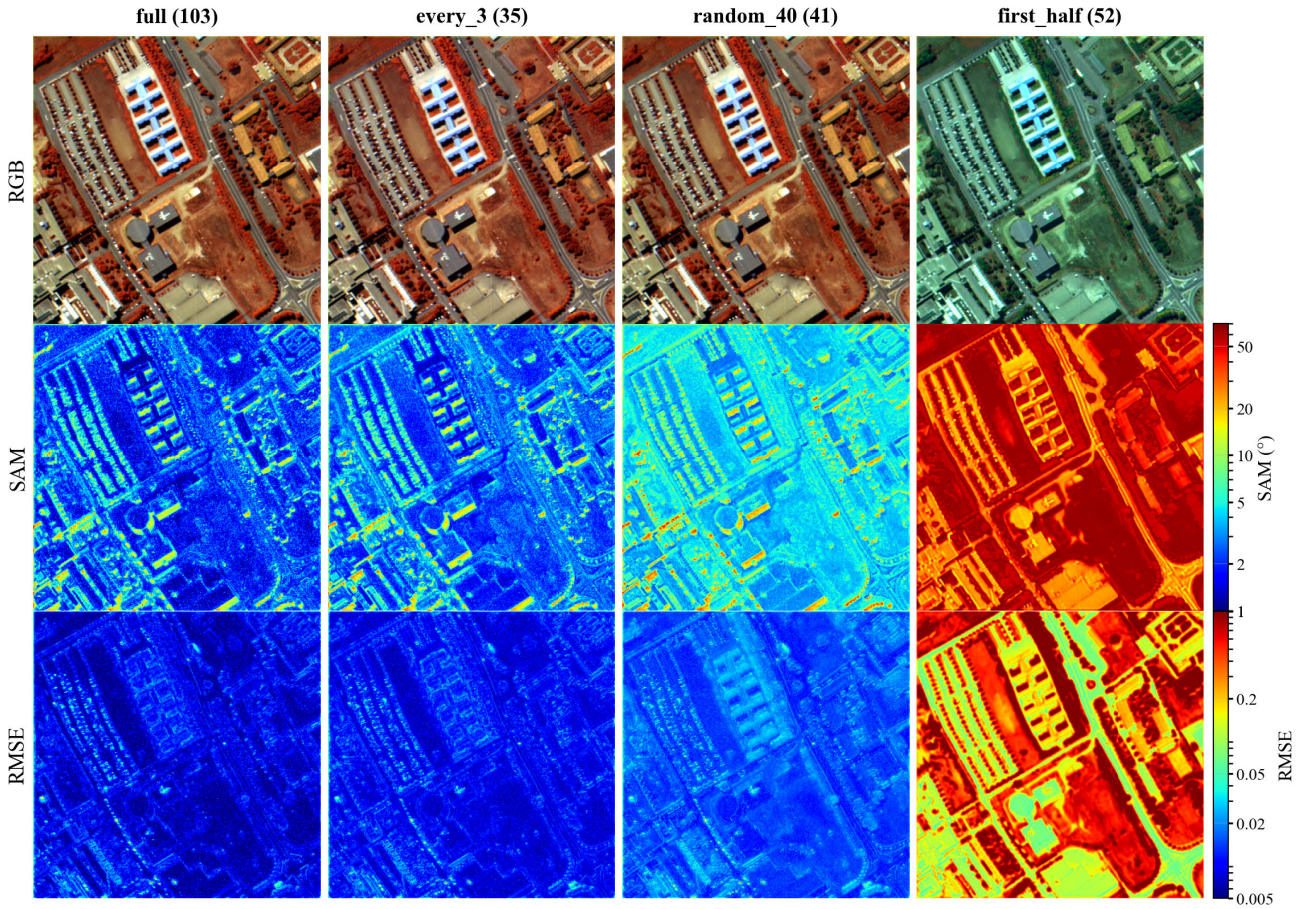


Fig. 13. Spatial visualization of cross-band fusion on PaviaU (full image, 336 × 336). (Top row) False-color composite (R/G/B = 718/594/490 nm, as in Fig. 5); (middle row) per-pixel SAM error map; and (bottom row) per-pixel RMSE error map. Because the first_half errors exceed those of the other configurations by more than an order of magnitude, both error rows use a logarithmic color scale shared across all columns. The first_half column (rightmost) shows the physical boundary collapse where uncovered NIR bands cause severe spectral and spatial errors.

TABLE VII

PHYSICAL WAVELENGTH EMBEDDING VERSUS CHANNEL-INDEX EMBEDDING ON PAVIAU (SCALE = 4, S2B, RANDOM-PATCH TRAINING WITH TILED FULL-IMAGE EVALUATION, FR = 4000). ONLY THE PER-BAND CONDITIONING EMBEDDING IS CHANGED; ALL OTHER COMPONENTS ARE HELD FIXED. FS: FROM-SCRATCH; PT: UNIVERSAL PRETRAINING ON THE LOO POOL {HOUSTON18, CHIKUSEI, TG} FOLLOWED BY UNSUPERVISED ADAPTATION. Δ_{PT-FS} ISOLATES THE BENEFIT CONTRIBUTED BY CROSS-SENSOR PRETRAINING. ALL ENTRIES ARE MEAN±STD OVER THE THREE SEEDS (42/7/123) OF TABLE II; THE WAVELENGTH ROWS ARE ITS OURS ROWS

	From-scratch		Pretrained		Δ_{PT-FS}	
Band encoding	PSNR	SAM	PSNR	SAM	PSNR	SAM
Physical wavelength (ours)	40.10±0.36	3.01±0.06	40.72±0.05	3.05±0.03	**+0.62**	+0.04
Channel index (learnable)	40.16±0.12	3.03±0.06	40.20±0.04	3.26±0.04	+0.04	+0.23

Two observations isolate the mechanism. *First, from scratch the two encodings are indistinguishable within run-to-run variance* (40.16 ± 0.12 versus 40.10 ± 0.36 dB): when a single fixed band layout is learned in isolation, an ordinal table can fit it as well as a wavelength encoding. *Second, the two diverge entirely under cross-sensor pretraining.* Physical-wavelength embedding converts the pretrained pool into a +0.62 dB mean gain (positive for every seed), whereas the channel-index variant gains nothing (+0.04 dB, with sign-inconsistent per-seed differences) and its spectral fidelity degrades in every seed (SAM 3.03 → 3.26). The reason is physical: the pool sensors (Houston18, Chikusei, and TG) place their bands at wavelengths and ordinal positions unrelated to PaviaU, so an index table trained on the pool addresses the wrong spectral content at adaptation time; a wavelength encoding, being sensor-independent, transfers directly. The convergence curves in Fig. 14 visualize this behavior: pretraining lifts the wavelength model onto a higher trajectory that persists to convergence, while for the index model the two curves collapse together: the pretrained warm-start is transient and fully decays. Physical-wavelength alignment is therefore the mechanism that enables cross-sensor pretraining to transfer, rather than a mere representational choice.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

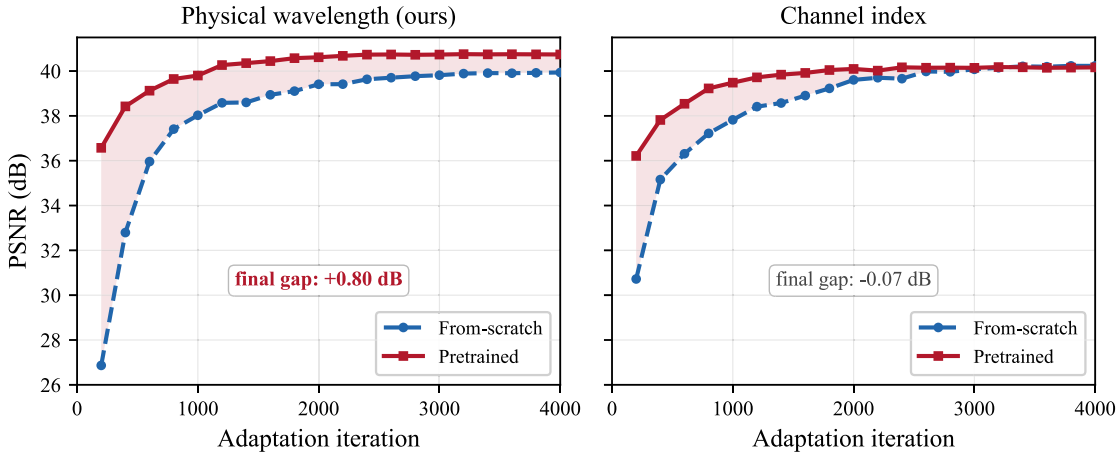


Fig. 14. Unsupervised adaptation convergence on PaviaU (Scale = 4, tiled full-image PSNR) under physical-wavelength (left) versus channel-index (right) band embedding. In each panel, the dashed blue curve is FS and the solid red curve is after cross-sensor pretraining; the shaded region is the pretraining gap. With physical-wavelength embedding the gap persists at ≈0.8 dB to convergence (seed-42 curves; + 0.62 dB in three-seed mean, Table VII); with channel-index embedding the warm-start decays and the two curves merge: cross-sensor pretraining transfers only when spectral reasoning is anchored to physical wavelength.

TABLE VIII

CROSS-SRF GENERALIZATION ON PAVIAU (SCALE = 4). LANDSAT-8 OLI HAS ONLY FOUR MSI BANDS WITH NONZERO RESPONSE WITHIN PAVIAU'S WAVELENGTH RANGE VERSUS S2B'S EIGHT BANDS. ALL ENTRIES ARE MEAN±STD OVER THREE RUNS UNDER THE TABLE II PROTOCOL (OURS OVER SEEDS 42/7/123, MIAE OVER ITS ORIGINAL-SEED RUN PLUS SEEDS 7/123); THE S2B ROWS REUSE THE TABLE II RUNS. BEST PER BLOCK IN BOLD, SECOND BEST UNDERLINED

SRF	MSI bands	Method	PSNR↑	SAM↓	ERGAS↓
S2B	8	MIAE	39.65±0.92	**2.95**±0.16	2.52±0.85
		Ours (FS)	40.10±0.36	3.01±0.06	1.93±0.11
		Ours (PT)	**40.72**±0.05	3.05±0.03	**1.90**±0.02
Landsat	4†	MIAE	38.64±0.43	3.72±0.13	2.19±0.06
		Ours (FS)	37.90±0.20	3.38±0.02	2.18±0.07
		Ours (PT)	**40.30**±0.20	**2.91**±0.04	**1.83**±0.06

†The OLI SRF product used here contains seven bands (blue, green, red, NIR, cirrus, SWIR-1, SWIR-2; the 443-nm coastal-aerosol and panchromatic bands are not included); 4 of them have nonzero spectral response within PaviaU's 430–838 nm range (response-weighted centers 483, 561, 655, and 865 nm; the NIR band contributes only through the lower tail of its response).

5) Cross-SRF Generalization: The final dimension is the SRF itself, which also tests whether a pretrained checkpoint survives a change of MSI sensor. Table VIII compares performance on PaviaU using the S2B SRF (eight MSI bands within 430–838 nm) versus the Landsat-8 Operational Land Imager (OLI) SRF; only 4 of the 7 bands in the OLI SRF product have nonzero spectral response within PaviaU's wavelength range, so the Landsat configuration provides substantially less spatial guidance and constitutes a harder fusion task. Note that the pretrained model used in the Landsat rows is the *same* checkpoint as in Table II; no repretraining is performed for the new SRF.

With only 4 Landsat MSI bands (versus S2B's 8), FS performance drops by −2.20 dB in three-run mean (37.90 versus 40.10), reflecting the reduced spatial guidance from fewer MSI bands; MIAE, whose low-rank spectral model leans less on the MSI, loses only −1.01 dB and overtakes our FS variant on this harder configuration. Pretraining, however, recovers nearly the entire loss: Landsat PT reaches 40.30 ± 0.20 dB, only −0.42 dB below S2B PT (40.72 dB), and outperforms both MIAE (+1.66 dB) and our own FS model (+2.40 dB) while also holding the best SAM and ERGAS of the block. The wavelength-aware architecture thus accommodates a different SRF without modification, and pretraining largely compensates for the reduced observation quality. The pretraining gain under Landsat (+2.40 dB in mean; +2.15 to +2.53 per seed) is roughly four times the S2B gain (+0.62 dB): pretraining is most valuable when the fusion task is harder.

E. Real Cross-Sensor Deployment on ZY1-02D

All experiments so far follow the Wald protocol on simulated observations. We finally test the framework on a *real* spaceborne observation pair acquired by the Chinese ZY1-02D satellite, the Liaoning-01 dataset of [41], for which no HrHSI reference exists, the exact deployment regime that motivates this work. The pair consists of an LrHSI of size 300 × 300 × 166 acquired by the Advanced Hyperspectral Imager (AHSI), covering 396–2501 nm and thus spanning both the visible–NIR (VNIR) and shortwave-infrared (SWIR) ranges, and a co-registered HrMSI of size 900 × 900 × 8 from the visible and NIR camera (VNIC), giving a scale factor $s = 3$. *Data preparation.* We use the co-registered pair with an exact 3× grid ratio as released in [41], where its acquisition and preprocessing are described; our only additional step is a relative radiometric normalization, in which the HrMSI is normalized to the LrHSI by per-band Theil–Sen lines fit over 15 020 pseudo-invariant pixels (mean $R^2 = 0.924$) between the LrHSI projected through the nominal VNIC SRF and the HrMSI Gaussian-blurred ($\sigma = 1.5$ HR pixels) and downsampled by 3. Both images are then clipped to $[0, 1]$ (affecting 0.12% of LrHSI and fewer than 0.001% of HrMSI values) without further stretching, and all 166 bands are retained; since the nominal VNIC SRF covers only 82 of them (up

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

TABLE IX

OBSERVATION CONSISTENCY ON THE REAL ZY1-02D PAIR (SCALE = 3, NO HRHSI REFERENCE, FULL-IMAGE TILED EVALUATION AT ITERATION 4000). OUR VALUES ARE MEAN±STD OVER THREE INDEPENDENT PAIRED RUNS (SEEDS 42/7/123). "SPECTRAL" PATH: $\hat{\mathbf{R}}\hat{\mathbf{X}}$ VERSUS HRMSI WITH EACH RUN'S OWN LEARNED SRF; "SPATIAL" PATH: $(\hat{\mathbf{X}} * k_{\hat{\theta}})\downarrow_s$ VERSUS LRHSI; "SPECTRAL, NOMINAL SRF": EVERY PREDICTION PROJECTED THROUGH THE SAME SENSOR-NOMINAL ZY1-02D SRF. THE PRETRAINED MODEL USES THE ALL-FOUR-DATASET CHECKPOINT, FOR WHICH ZY1-02D IS AN UNSEEN SENSOR

Path	Method	PSNR↑	SAM↓	ERGAS↓	SSIM↑
Spectral	From-scratch	40.34±1.91	2.34±0.47	3.48±0.66	0.961±0.040
	Pretrain	**45.52**±0.47	**1.48**±0.03	**2.47**±0.03	**0.991**±0.000
Spatial	From-scratch	41.13±0.21	**4.12**±0.66	**3.19**±0.10	0.974±0.001
	Pretrain	**41.13**±0.22	4.36±0.63	3.21±0.06	**0.975**±0.002
Spectral, nominal SRF	CNMF*	39.85	3.53	3.69	0.972
	MIAE	33.69	6.40	8.20	0.934
	From-scratch	38.85±1.45	3.72±0.56	3.86±0.59	0.954±0.040
	Pretrain	**41.63**±0.09	**3.02**±0.22	**3.02**±0.05	**0.984**±0.001

CNMF and MIAE are single runs evaluated at their final iterate. *CNMF receives the nominal SRF as an input (Section IV-A), so its consistency under this operator is fitted by construction.

to 1089 nm), the SWIR bands are constrained by the spatial path alone. Stage 1 learns the unknown ZY1-02D degradation directly from this pair, and the fusion network is adapted with the identical merged objective used throughout.

For the pretrained variant, we reuse a single checkpoint pretrained on *all four* simulated benchmarks (PaviaU, Houston18, Chikusei, and TG) with the S2B SRF at 4×; ZY1-02D shares neither its sensor, band layout (166 bands to 2501 nm), SRF, nor scale factor with any pretraining source, so this constitutes an unseen-sensor transfer. The pretrained weights load directly into this configuration without any architectural change, so the band-agnostic design is structurally portable across the simulated-airborne-to-real-spaceborne gap. The SWIR portion of the target range (beyond ~1030 nm) lies entirely outside the wavelength span of the pretraining pool (363–1033 nm), so the wavelength encoding is queried far beyond its pretraining support; the results below indicate that per-scene adaptation absorbs this extrapolation.

1) Observation-Consistency Evaluation: Without an HrHSI reference, standard reference metrics are undefined. We therefore quantify *observation consistency*: how faithfully the fused $\hat{\mathbf{X}}$ reproduces the two real observations after reapplying the learned degradation operators. Along the spatial path we compare $(\hat{\mathbf{X}} * k_{\hat{\theta}})\downarrow_s$ against the LrHSI $\mathbf{Y}$ (on the observed HSI bands), and along the spectral path we compare $\hat{\mathbf{R}}\hat{\mathbf{X}}$ against the HrMSI $\mathbf{Z}$, reporting PSNR, SAM, ERGAS, and SSIM for each path. High spectral-path consistency indicates that the reconstruction respects the sensor's true spectral sampling, while spatial-path consistency reflects agreement with the observed coarse HSI. Table IX reports the final values and Fig. 15 the full adaptation trajectories.

2) Results: On this real unseen sensor, universal pretraining transfers effectively and yields a substantial spectral-path improvement: + 5.18-dB PSNR in three-run mean (45.52 ± 0.47 versus 40.34 ± 1.91), with SAM reduced from 2.34° to 1.48° and SSIM raised to 0.991; the spatial-path consistency is essentially unchanged, so the gain concentrates on the harder spectral-reconstruction subproblem. The variance pattern mirrors the simulated benchmarks: one FS run drifts to a visibly worse spectral solution (38.14 dB), whereas no pretrained run deviates from the mean by more than 0.5 dB, a valuable property on a real sensor, where no reference is available to detect a degraded solution. The trajectories in Fig. 15 likewise mirror Section IV-C: the pretrained model reaches near-peak spectral SSIM (≈ 0.985) within a few hundred iterations, a quality the FS model attains only after roughly 2000 iterations, an order-of-magnitude speedup, larger than the 2.0–3.3× of Section IV-C because consistency SSIM saturates earlier than PSNR and the warm-start advantage grows on a fully unseen sensor. Because no HrHSI supervision is used at any point, this experiment demonstrates the complete "train once, fuse anywhere" pipeline on a real cross-sensor pair.

Fusion Quality Versus Observation Fitting: Since the upper spectral block scores each run with its own Stage-1 SRF, a gain could partly reflect a tighter fit to that estimate. We separate the two effects in two ways. First, Stage 1 does not depend on pretraining, so paired FS and pretrained runs share bit-identical Stage-1 operators, and scoring all six runs with one common learned SRF (seed 42) still gives +4.93 dB. Second, under the common sensor-nominal SRF (lower block), the pretrained model remains the best on all metrics and on every seed (41.63 ± 0.09 versus 38.85 ± 1.45 dB, +2.79 dB). The smaller margin partly reflects the weaker fit of the nominal SRF to this normalized pair: without any fusion, the LrHSI mapped through it matches the 3 × 3 block-averaged HrMSI at 34.95 dB, versus 35.21 dB with the learned SRF. CNMF, which is given the nominal SRF and thus fit to it by construction, slightly exceeds our FS mean (39.85 dB) but not the pretrained model; MIAE trails both (33.69 dB).

3) Cross-Method Comparison via No-Reference Metrics: Observation consistency measured with each method's *own* learned operators cannot compare different methods, and the common nominal-SRF block of Table IX covers only the spectral path. As a complementary cross-method comparison on this real pair we use the reference-free QNR index [81], which depends only on the fused image and the two observations: it combines a spectral-distortion term D_λ (inconsistency of interband Q relationships between the fused image and the LrHSI) and a spatial-distortion term D_s (inconsistency of the fused image with a synthetic panchromatic band formed from the HrMSI), with QNR = $(1 - D_\lambda)(1 - D_s)$. We run two representative baselines on the identical pair: CNMF, the classical coupled-factorization method, and MIAE, the strongest unsupervised baseline of Table II; since no reference exists for checkpoint selection, each method is evaluated at its final iterate. The two most recent DIP-based baselines could not be included: the official CS2DIPs implementation does not scale to the 900 × 900 × 166 target (its coupled generator fails at this size), and EDIP-Net's official implementation takes a reference image as a required input (to size its output and to report metrics inside its loop), which we did not adapt to the reference-free setting. Table X reports the results, computed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

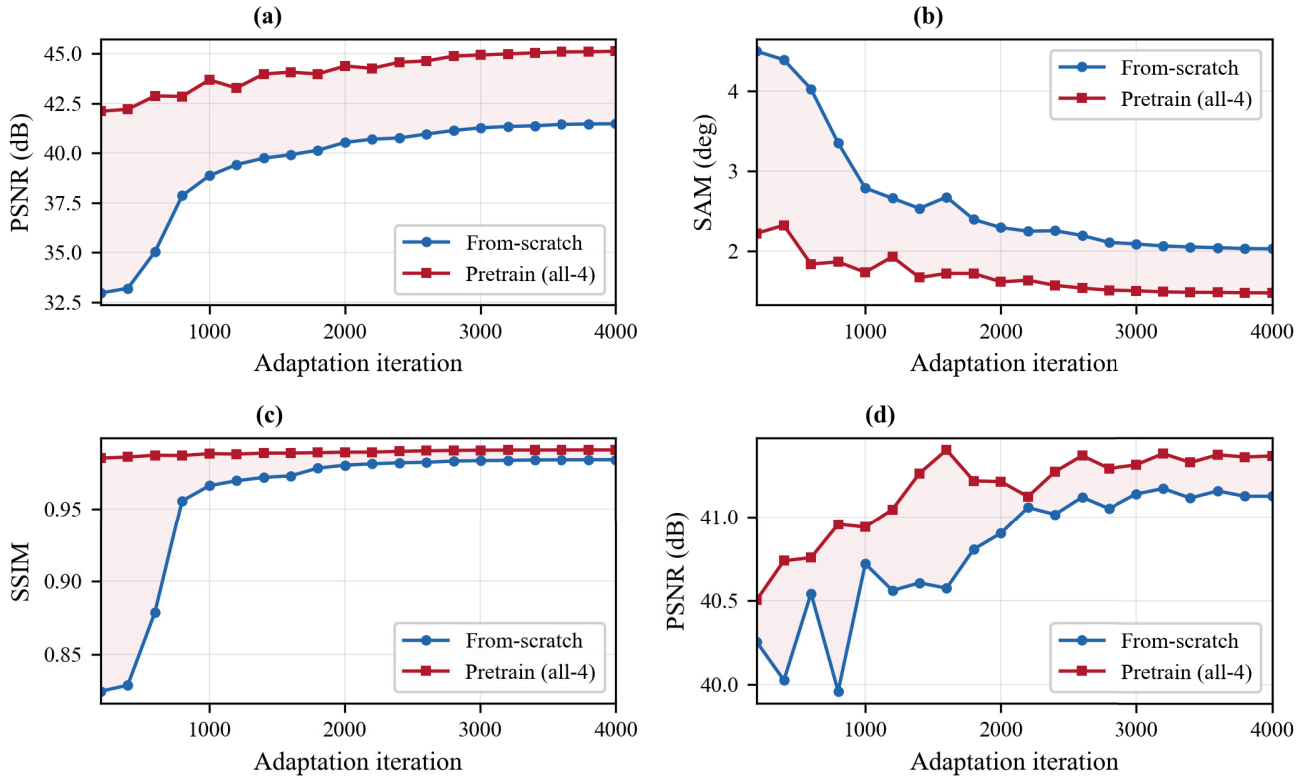


Fig. 15. Unsupervised adaptation on the real ZY1-02D pair (observation consistency, full-image tiled evaluation every 200 iterations; no HrHSI reference). Blue: FS; red: pretrain on all four simulated benchmarks (ZY1-02D unseen). Shaded region marks the pretraining advantage. (a)–(c) Spectral-path consistency ($\hat{\mathbf{R}}\hat{\mathbf{X}}$ versus HrMSI) in PSNR, SAM, and SSIM; (d) spatial-path PSNR $[(\hat{\mathbf{X}} * k_{\hat{\theta}})\downarrow_s$ versus LrHSI]. Curves show the seed-42 paired run; Table IX reports three-run statistics. The pretrained model reaches near-peak spectral consistency within a few hundred iterations, whereas FS needs roughly an order of magnitude more.

TABLE X

REFERENCE-FREE QNR COMPARISON ON THE REAL ZY1-02D PAIR. D_λ: SPECTRAL DISTORTION; D_s: SPATIAL DISTORTION; QNR = $(1-D_\lambda)(1-D_s)$. COMPUTED IDENTICALLY FOR ALL METHODS FROM THE FUSED IMAGE AND THE TWO OBSERVATIONS ONLY, INDEPENDENT OF ANY LEARNED DEGRADATION OPERATORS. BEST IN BOLD, SECOND BEST UNDERLINED

Method	$D_\lambda \downarrow$	$D_s \downarrow$	QNR↑
CNMF	0.0287	0.0751	0.8983
MIAE	0.0438	0.0455	0.9127
Ours (FS)	<u>0.0183</u>	<u>0.0265</u>	<u>0.9557</u>
Ours (PT)	**0.0171**	**0.0264**	**0.9569**

with a $32 \times 32Q$ window over 40 uniformly sampled bands on the seed-42 paired run.

Both baselines trail by a wide margin. MIAE's spectral distortion is 2.6× ours (D_λ0.0438 versus 0.0171), visible in Fig. 16 as a pronounced reddish color cast over water and vegetation, and CNMF's spatial distortion is 2.8× ours (D_s0.0751 versus 0.0264), consistent with its over-smoothed reconstruction. Between our two variants, pretraining lowers the spectral distortion (D_λ: 0.0183 → 0.0171) while leaving the spatial distortion unchanged, reproducing, under a metric independent of our learned degradation operators, the finding that the pretraining gain is concentrated in spectral fidelity. Fig. 16 presents the corresponding false-color composites: our fused results restore the fine spatial structure of the HrMSI, absent in the coarse LrHSI, while carrying the full 166-band spectral content and preserving natural colors; FS and pretrained reconstructions are visually close, as the quantitative gain lies in spectral accuracy rather than in the three displayed bands.

F. Ablation Study

This section ablates the three design levels of AnyFusion: the merged training strategy, the SST architectural components, and the joint degradation learning of Stage 1. All ablations follow the production protocol of the main comparison (random-patch training, tiled full-image evaluation).

1) Merged Versus Sequential Training: We first evaluate the merged RR+FR objective of (18) against the sequential baseline (RR pretraining followed by FR optimization) on PaviaU.

As shown in Table XI, merged training improves PSNR by +1.04 dB over the sequential schedule at only 10% additional wall-clock time. The RR branch anchors the network with direct self-generated supervision, while the FR branch optimizes the actual HR domain; optimizing them jointly avoids the regime shift of a hard stage switch and yields a learning signal that neither branch provides alone. The merged strategy is therefore adopted in all other experiments.

2) SST Component Ablation: We next ablate the SST backbone components introduced in Section III-C on PaviaU

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

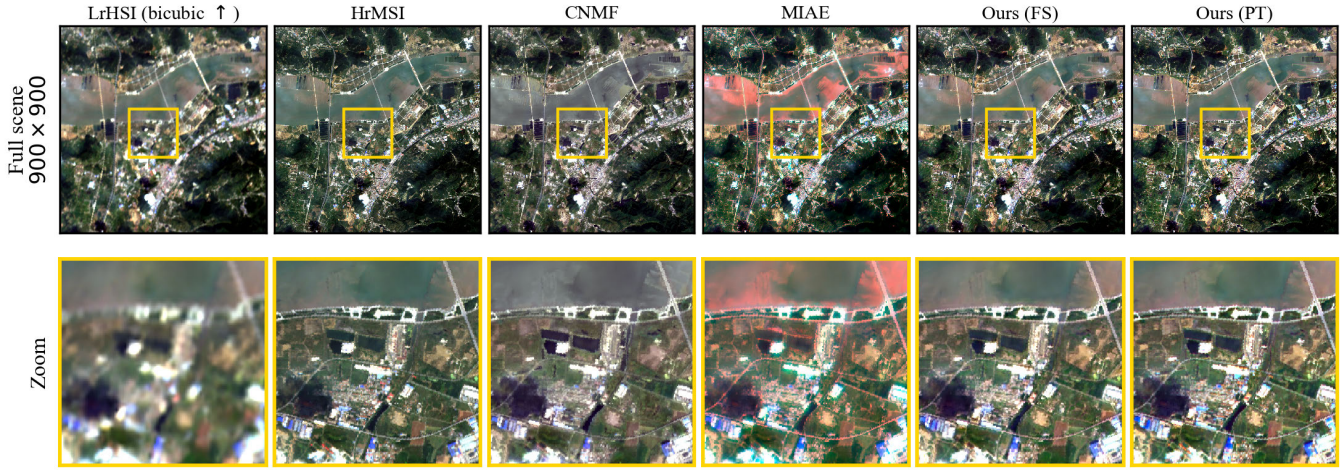


Fig. 16. Qualitative fusion on the real ZY1-02D pair (false-color composite of the bands nearest to 640/550/460 nm, i.e., 636/550/456 nm; no HrHSI reference). (Top) Full 900 × 900 scene with the zoom region boxed; (bottom) the corresponding zoomed-in view. (Left to right) Bicubic-upsampled LrHSI, HrMSI, CNMF, MIAE, Ours (FS), and Ours (pretrain). All HSI-derived panels share an identical percentile stretch computed from the LrHSI, so colors are directly comparable across methods. MIAE exhibits a pronounced reddish spectral cast and CNMF over-smooths water regions, whereas our fused results recover the fine spatial structure of the HrMSI while preserving natural colors.

TABLE XI

TRAINING STRATEGY ABLATION ON PAVIAU (FS, SCALE = 4, RANDOM-PATCH 256, TILED FULL-IMAGE EVALUATION). BOTH STRATEGIES WERE TRAINED AS A CONTROLLED SAME-BATCH PAIR; THE DIFFERENCE OF THE MERGED ROW FROM THE SEED-42 REFERENCE RUN OF TABLE II (39.94 DB) IS SINGLE-RUN VARIANCE

Strategy	PSNR↑	SAM↓	ERGAS↓	Time (s)
Separate RR→FR	39.34	3.71	2.33	3989
Merge ($\lambda_{rr} = 1.0$)	**40.38**	**3.01**	**1.97**	4407
Δ	+1.04	−0.70	−0.36	+10%

TABLE XII

SST ARCHITECTURE ABLATION ON PAVIAU (FS, SCALE = 4; RANDOM-PATCH (256) TRAINING WITH TILED FULL-IMAGE EVALUATION, FR = 4000). EACH ROW REMOVES OR REPLACES ONE COMPONENT FROM THE FULL SST BASELINE. ALL ENTRIES ARE MEAN±STD OVER SEEDS 42/7/123; THE BASELINE ROW IS THE FS ENTRY OF TABLE II, AND Δ IS THE DIFFERENCE OF PSNR MEANS

Variant	PSNR↑	SAM↓	Δ dB
SST baseline	40.10±0.36	3.01±0.06	—
w/o SRF-overlap bias	39.17±0.62	3.25±0.14	−0.93
w/o $\Delta\lambda$ bias	40.09±0.24	3.02±0.06	−0.01
Softmax pool (replace multi-head)	38.59±1.75	3.27±0.37	−1.51
w/o FiLM conditioning	39.13±0.92	3.23±0.16	−0.97
$n_b = 2$ (shallow)	39.90±0.36	3.03±0.06	−0.20

(scale = 4, FS) under the production protocol (FR = 4000); the full network has 3.50-M parameters. Because single-run deltas can be comparable to the ±0.36 dB seed std of the FS baseline (see Table II), every variant is run over the same three seeds (42/7/123) as the main comparison, and Table XII reports mean±std throughout.

a) Multihead attention pooling: It carries the largest contribution: replacing it with single-head softmax pooling costs −1.51 dB in mean, with one seed collapsing to 36.57 dB (std 1.75 versus the baseline's 0.36). The rank-one bottleneck is thus not only an accuracy cost but a stability risk, confirming that diverse spectral aggregation heads are needed to summarize the band-token set without collapsing it to a single convex combination.

b) SRF-overlap bias: (−0.93 dB) and **FiLM conditioning** (−0.97 dB) contribute comparably, both with degradations well beyond the baseline seed std and with SAM rising in parallel: the former confirms that encoding the physical SRF overlap between MSI and HSI bands provides meaningful guidance for the spectral set attention, and the latter that the wavelength-conditioned modulation of the shared band-lifting and spatial-mixing operators cannot be compensated by the attention biases alone. We note that the FiLM row is a case where multiseed evaluation overturns a single-run reading: its seed-42 run (40.19 dB, nominally + 0.25 above the seed-42 baseline) turns out to be the favorable outlier among its three seeds (38.70 and 38.51 dB for the other two).

c) δλ relative bias: It is the one attention-side component whose removal causes no degradation (−0.01 dB in mean, stable across seeds): with FiLM active, the wavelength information it encodes is already available to every token, so the remaining mechanism compensates. Its physical grounding matters nonetheless through its shared input: replacing the physical wavelength embedding itself by a channel-index table eliminates the cross-sensor pretraining gain entirely (see Table VII).

d) Network depth: (n_b) is a secondary factor at the full training budget: the shallow $n_b = 2$ variant trails the baseline by only −0.20 dB in mean (within the seed std). We retain $n_b = 4$, which all pretrained checkpoints use.

3) Learned Versus Oracle Degradation: Finally, we ask whether the Stage 1 estimates limit reconstruction quality, i.e., whether fusion would improve with the *true* PSF and SRF. A fair oracle must match both the true kernel shape and the sampling geometry of the data: our simulation (see Section IV-A) keeps every sth HR pixel after blurring (LR pixel i centered on HR pixel si), whereas the operators of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

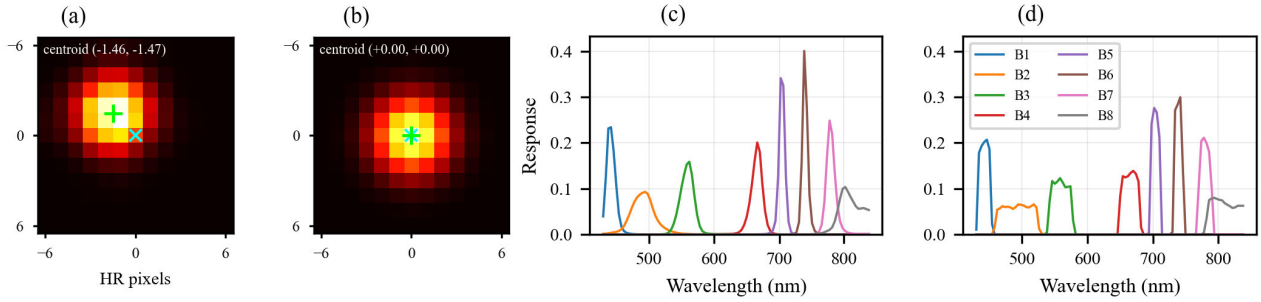


Fig. 17. Learned versus oracle degradation operators on PaviaU (seed 42). (a) Learned PSF and (b) oracle Gaussian PSF on the same 13×13 HR grid; green +: kernel centroid, cyan $\times$: grid center. The learned kernel has the width of the oracle but is shifted by $(-1.46, -1.47)$ HR pixels, recovering the sampling-phase offset $-(s-1)/2 = -1.5$ between the simulated data and the resampling of Section III-B. (c) Learned SRF and (d) nominal S2B SRF of the eight S2B bands retained on PaviaU, sampled at the HSI band centers.

TABLE XIII

LEARNED VERSUS FIXED GROUND-TRUTH DEGRADATION OPERATORS ON PAVIAU (FS, SCALE = 4; RANDOM-PATCH TRAINING WITH TILED FULL-IMAGE EVALUATION, FR = 4000). "ALIGNED" FIXED OPERATORS ARE APPLIED WITH THE SAMPLING PHASE OF THE SIMULATED DATA; "OFFSET" ONES WITH THE BLOCK-CENTER RESAMPLING OF SECTION III-B, I.E., WITH A 1.5-HR-PIXEL PHASE OFFSET. TOP BLOCK: WALD-PROTOCOL GAUSSIAN SYNTHESIS; THE LEARNED AND ALIGNED-ORACLE ROWS ARE MEAN±STD OVER SEEDS 42/7/123 (THE LEARNED ROW IS THE FS ENTRY OF TABLE II), THE OFFSET ROW IS SEED 42. LOWER BLOCKS (SEED 42): THE LRHSI IS RESYNTHESIZED WITH EACH NON-GAUSSIAN PSF; "EXACT PSF" FIXES THE GROUND-TRUTH KERNEL AND "ASSUMED GAUSSIAN" A WALD GAUSSIAN, BOTH ALIGNED AND GRANTED THE GROUND-TRUTH SRF. BEST PER BLOCK IN BOLD

True PSF	Degradation	PSNR↑	SAM↓	ERGAS↓
Gaussian (Wald)	Learned INR	**40.10**±0.36	**3.01**±0.06	**1.93**±0.11
	Oracle PSF + SRF, aligned	39.74±0.52	3.11±0.07	2.05±0.26
	Oracle PSF + SRF, offset	37.20	4.20	2.85
Motion 45°	Learned INR	39.51	3.21	2.12
	Exact PSF	**39.84**	**3.03**	**1.97**
	Assumed Gaussian	38.87	3.38	2.40
Anisotropic	Learned INR	**40.02**	**3.11**	**2.02**
	Exact PSF	38.99	3.28	2.37
	Assumed Gaussian	38.80	3.36	2.42
Bimodal	Learned INR	**39.94**	**3.07**	**1.96**
	Exact PSF	39.01	3.22	2.39
	Assumed Gaussian	38.26	3.63	2.58

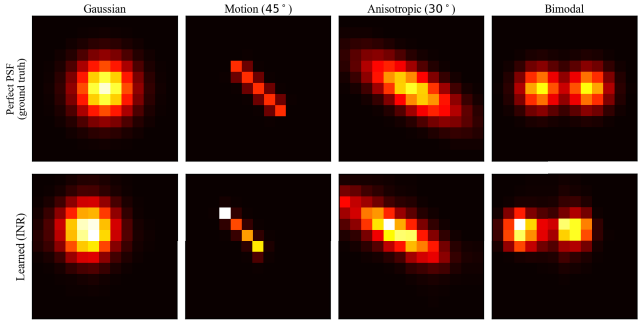


Fig. 18. Learned versus exact PSF on PaviaU. (Top) The exact ground-truth PSF used to synthesize the LrHSI (the "Exact PSF" setting of Table XIII) (Gaussian, 45° motion, anisotropic 30°, bimodal). (Bottom) The kernel learned by the Stage-1 INR from the observations alone. Despite a Gaussian smoothness prior, the INR captures the streak, anisotropy, and bimodality of each true operator; every learned kernel is displaced toward the upper left by the sampling-phase offset of 1.5 HR pixels discussed in the text. The leftmost (Gaussian) column is the non-Gaussian baseline, analogous to Fig. 17(a) and (b).

Section III-B resample at pixel-block centers $(si + (s-1)/2)$. The INR PSF, which can place its mass anywhere within its support, recovers this offset from the observations alone: its centroid lies at $(-1.46, -1.47)$ HR pixels on PaviaU [see Fig. 17(a)] and, at all seven scales of Table IV, matches the theoretical offset ($-(s-1)/2$ on integer grids, zero on fractional ones) to within 0.03 HR pixels, while its width about the centroid is that of the oracle kernel (standard deviation 1.64 versus 1.70 HR pixels).

The top block of Table XIII therefore compares the learned operators with oracle operators applied with the matching sampling phase and with a 1.5-HR-pixel (0.375-LR-pixel) phase offset, everything else being identical. Phase-matched oracle operators reach 39.74 ± 0.52 dB, comparable to the learned ones (40.10 ± 0.36 dB; the paired per-seed differences of 0.10–0.72 dB, all in favor of the learned operators, lie within the seed variance): joint learning attains oracle-level fusion accuracy without knowledge of the kernel or of the sampling geometry. With the phase offset, the same exact operators lose 2.01 dB (seed 42: 37.20 versus 39.21 dB). Since the relative sampling phase of real LrHSI and HrMSI grids is rarely known to subpixel accuracy, this sensitivity of fixed operators is where estimating the degradation from the observations is most useful. The pretraining losses of Section III-E evaluate the nominal operators with this phase offset; adaptation re-estimates the degradation on the target scene, so the offset does not reach deployment. The learned SRF [see Fig. 17(c)] preserves the bandpass structure of the nominal S2B SRF (d) with smoother band edges.

4) Robustness to Non-Gaussian Degradation: The INR PSF assumes neither a Gaussian nor a box-shaped kernel (see Section III-B). We test this by resynthesizing the PaviaU LrHSI with three non-Gaussian PSFs, a 45° motion blur, an anisotropic Gaussian (axis ratio ≈ 2.7:1, rotated 30°), and a bimodal kernel (see Fig. 18, top), and compare our learned degradation with two fixed, phase-matched PSFs that are granted the ground-truth SRF: the exact synthesis kernel ("exact PSF") and a Wald Gaussian ("assumed Gaussian").

The lower blocks of Table XIII (single runs; differences below the 0.36–0.52 dB seed std of the top block should not be over-interpreted) show that the learned pipeline is essentially

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

TABLE XIV

SENSITIVITY TO OBSERVATION NOISE (SCALE = 4; RANDOM-PATCH TRAINING WITH TILED FULL-IMAGE EVALUATION). ALL ROWS ARE SINGLE RUNS UNDER THE PROTOCOL OF TABLE II; THE $\sigma = 0$ ROWS ARE THE CORRESPONDING CLEAN-OBSERVATION RUNS, SO THAT Δ ISOLATES THE EFFECT OF NOISE FROM SEED VARIANCE. GAUSSIAN NOISE WITH STANDARD DEVIATION σ IS ADDED TO THE HRMSI OBSERVATIONS BEFORE TRAINING (CLIPPED TO $[0, 1]$); THE REFERENCE REMAINS CLEAN. MIAE IS EVALUATED UNDER THE IDENTICAL PROTOCOL. Δ IS THE PSNR CHANGE RELATIVE TO EACH METHOD'S OWN $\sigma = 0$ RESULT. FOR EACH $\sigma > 0$, THE BETTER PSNR AND SAM PER DATASET ARE IN BOLD

		PaviaU (103b)			**Houston18 (46b)**			**Chikusei (128b)**			**TG (54b)**		
σ	**Method**	PSNR	SAM	Δ	PSNR	SAM	Δ	PSNR	SAM	Δ	PSNR	SAM	Δ
0.00	Ours (FS)	39.94	3.04	—	44.96	1.77	—	47.18	1.61	—	43.44	1.23	—
	MIAE	40.22	2.89	—	44.65	2.46	—	49.67	2.52	—	43.58	1.22	—
0.01	Ours (FS)	**38.67**	**3.47**	−1.27	**42.90**	**2.21**	−2.06	**43.79**	**2.12**	−3.39	**40.66**	**1.58**	−2.78
	MIAE	37.43	4.52	−2.79	40.50	3.89	−4.15	42.93	4.09	−6.74	39.06	2.20	−4.52
0.05	Ours (FS)	**28.47**	**11.41**	−11.47	**30.37**	**11.43**	−14.59	31.49	**11.06**	−15.69	**27.50**	**7.96**	−15.94
	MIAE	25.84	26.38	−14.38	27.00	24.84	−17.65	**34.91**	25.04	−14.76	27.00	18.59	−16.58

Our $\sigma = 0$ entries are the seed-42 reference runs of the Table III protocol; MIAE's are its single clean reference runs, which differ from the three-run means of Table III owing to its run-to-run variance (most visibly on Chikusei: 49.67 vs. 48.81 ± 0.37); our seed-42 runs likewise differ from our means within the from-scratch seed std (most visibly on Chikusei: 47.18 vs. 45.84 ± 1.72). Anchoring each Δ to the method's own clean run isolates the noise effect; re-anchoring MIAE's Δ to the Table III means would change it by at most 0.86 dB without altering any comparison.

invariant to the PSF shape (39.5–40.0 dB). For fixed operators the shape does matter: the exact PSF beats the assumed Gaussian in every case (+0.96, +0.19, and +0.74 dB for motion, anisotropic, and bimodal blur). The learned operators stay within about 1 dB of the exact ones (slightly below for motion blur, 39.51 versus 39.84 dB, and above for the other two kernels) and beat the assumed Gaussian by 0.64–1.68 dB. The sampling phase again outweighs the kernel shape: with the 1.5-HR-pixel offset, the exact kernels reach only 36.98–37.13 dB, 1.88–2.86 dB below their phase-matched counterparts. The learned kernels (see Fig. 18, bottom) reproduce the structure of each operator, displaced by the same phase offset.

G. Sensitivity to Observation Noise

The analyses so far assume clean observations. Since unsupervised fusion couples the reconstruction directly to the observations through consistency losses, observation noise deserves explicit quantification on both observation paths. Table XIV evaluates the proposed method (FS) with additive Gaussian noise of standard deviation σ injected into the HrMSI, alongside MIAE, the strongest unsupervised baseline of Table II, under the identical noise protocol; noise on the LrHSI side is examined at the end of this section.

At moderate noise ($\sigma = 0.01$), the proposed method degrades gracefully (−1.27 to −3.39 dB) and attains both higher PSNR and lower SAM than MIAE on all four datasets, including PaviaU and Chikusei, where MIAE's reference runs in this table hold the higher clean-observation PSNR. MIAE loses 2.79–6.74 dB at the same operating point, 1.6–2.2× our degradation. At severe noise ($\sigma = 0.05$), both methods degrade substantially (−11.5 to −15.9 dB for ours, −14.4 to −17.7 dB for MIAE), and MIAE additionally collapses spectrally (SAM 18.6°–26.4° versus our 8.0°–11.4°); its nominally higher PSNR on Chikusei (34.91 dB) comes at a 25.04° SAM, i.e., spatial smoothing at the expense of spectral fidelity. Together these results indicate that sensitivity to observation noise is a systemic property of unsupervised observation-consistency fusion (without ground-truth supervision, the consistency losses propagate noise into the reconstruction for any method of this class) rather than a weakness specific to our design; within the paradigm, the proposed method is the more tolerant to MSI-side noise.

Noise on the LrHSI Side: Repeating the protocol on PaviaU with the noise injected into the LrHSI instead, the sensitivity pattern *inverts*: MIAE is nearly unaffected at $\sigma_{\text{hsi}} = 0.01$ (40.18, −0.04 dB) and loses 4.41 dB at $\sigma_{\text{hsi}} = 0.05$, whereas our method drops by 2.25 and 12.51 dB (37.69 and 27.43 dB), with SAM rising to 13.6° versus MIAE's 4.9°. The two noise axes are thus complementary, and each method's inductive bias determines which observation it is anchored to: MIAE's compact endmember bottleneck acts as an implicit spectral denoiser for LrHSI noise but propagates HrMSI noise into the spectra (SAM 18.6°–26.4° above), while our reconstruction couples to the LrHSI through the interpolated spectral residual (15) and the RR supervision (17), both of which pass LrHSI noise through. Neither method dominates both axes. The real ZY1-02D pair of Section IV-E, whose AHSI observation carries real sensor noise, is nonetheless fused successfully; the synthetic perturbation $\sigma = 0.05$ on a $[0, 1]$ reflectance scale overstates typical deployment conditions. Explicit noise modeling in the unsupervised setting remains future work.

H. Computational Efficiency

We close with the efficiency dimension raised in Section I. We do not claim that an AnyFusion iteration is cheap (it is not; see below), but that universal pretraining changes *how the cost scales with deployment*: the expensive part is paid once, each adaptation needs 2.0–3.3× fewer iterations, and the outcome becomes reproducible without a reference. Table XV reports, on one RTX 3090, the parameter count, the iteration budget of each method's official configuration, the floating-point operations (FLOPs), and the measured rate in seconds per 1000 iterations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

TABLE XV

COMPUTATIONAL EFFICIENCY COMPARISON ON A SINGLE NVIDIA RTX 3090 GPU. "FLOPS" IS THE FORWARD COMPUTATION OF ONE OPTIMIZATION ITERATION ON PAVIAU, MEASURED WITH THE PYTORCH PROFILER UNDER A COMMON PROTOCOL (MULTIPLY–ADD = 2 FLOPS; BACKWARD PASSES EXCLUDED). "RATE" IS THE MEASURED WALL-CLOCK TIME PER 1000 ITERATIONS OF EACH METHOD'S FUSION-OPTIMIZATION LOOP (PERIODIC IN-LOOP EVALUATION INCLUDED), OBTAINED BY NORMALIZING THE LOOP'S WALL-CLOCK TIME BY ITS ITERATION BUDGET "ITERS"; END-TO-END TIME-TO-RESULT IS THEIR PRODUCT (+~40-S STAGE-1 DEGRADATION LEARNING FOR OURS). "PARAMS" IS THE NUMBER OF LEARNABLE PARAMETERS. CNMF RUNS MULTIPLICATIVE UPDATES ON CPU (FLOPS COUNTED ANALYTICALLY)

Type	Method	Params	Iters	GFLOPs /iter.	Rate (s/1000 iter.) PaviaU	Hou.	Chik.	TG
Trad.	CNMF	—	0.3k	1.3	16	4	67	4
Unsup. DL	CUCaNet	~1.2M	8k	327	53	27	141	27
	FusionNet	~0.8M	10k	276	43	20	104	20
	UDALN	~2.5M	15k	323	37	18	92	18
	DTDNML	~1.0M	10k	48	15	7	36	7
	MIAE	~3.0M	10k	12	11	9	25	9
	CS2DIPs	~9.9M	5k	212	137	87	274	89
	EDIP-Net	~12.9M	22k†	836	176	72	542	76
Ours	from-scratch	3.50M	4k	872	927	282	654	327
	pretrain adapt	3.50M	4k	872	925	280	654	326

Note: iterations are not commensurable units of work across methods; the baselines optimize on the full image (so their FLOPs and rate grow with scene size), whereas ours optimizes on fixed random patches (256^2 on PaviaU, 192^2 elsewhere) with two forward–backward passes (RR and FR) per iteration; both forward passes are included in its FLOPs (backward excluded, as for all methods), and both full forward–backward passes in its rate. Our rates are re-measured single runs of the production configuration in one controlled batch and exclude the one-off Stage-1 degradation learning (~40 s) and the final tiled inference ($\leq$4 s). †Four-stage pipeline: three optimized stages (blind degradation estimation, spectral up-sampling, and DIP optimization; $2 \times (2000 + 2000 + 7000)$ iterations including the equal-length decay schedule of each stage) followed by a training-free decision-fusion stage; FLOPs and rate are averaged over the optimized stages. "Params" of our method counts the complete fusion network (backbone 2.44M, wavelength-query decoder 0.86M, SRF embedding 0.20M); the Stage-1 degradation MLPs (~28k) are excluded.

Because learning rates, schedules, and losses differ, iteration budgets (300–22 000) are not comparable convergence milestones; the FLOPs and rate columns are the per-step comparisons, and end-to-end cost is Rate $\times$ Iters/1000. Per iteration our method is the most expensive (872 GFLOPs and 927 s per 1000 iterations versus 12 GFLOPs and 11 s for MIAE on PaviaU), as each iteration runs RR and FR forward–backward passes of the band-token network. Parameter count predicts neither FLOPs nor rate: our 3.5-M-parameter model costs 73$\times$ the FLOPs of the 3.0-M-parameter MIAE, because shared weights are reapplied per band and per queried wavelength, and at nearly identical FLOPs (872 versus 836) our rate is 5.3$\times$ EDIP-Net's on PaviaU. This gap is the cost of the band-token representation: every backbone operator streams a $[N, d_b, H, W]$ activation tensor (~233 MB at half precision on PaviaU) through GPU memory, so the iteration is bound by memory bandwidth rather than arithmetic. Two effects qualify these per-step costs: the baselines iterate on the full image, so their cost grows with scene size (EDIP-Net: 176 on PaviaU to 542 on Chikusei, approaching our patch-based rate), and our budget of 4000 iterations is the smallest among the learned methods. End to end, the method is on the expensive side of the paradigm (~62 min on PaviaU, ~19–23 min on the 240^2 scenes) but not its extreme (Chikusei: 2660 s versus EDIP-Net's 11921 s).

Pretraining leaves the rate unchanged (within 1% on all four datasets); it reduces the number of iterations, not their cost, through three mechanisms.

1) Amortization: The one-time pretraining (10k steps on three datasets, ~2 h) yields a checkpoint reusable for any new sensor configuration.

2) Equal-Quality Budgets: Matching the FS final quality takes 2.0–3.3$\times$ fewer iterations (see Table III), shrinking the wall-clock time by the same factor.

3) Unsupervised Early Stopping: Since quality increases monotonically with diminishing returns (see Fig. 11), a reference-free rule (stop once the exponential moving average of the training loss improves by less than 0.5% over 500 iterations) stops the four pretrained adaptations at iterations 2400–3800, within 0.06 dB of the 4000-iteration PSNR on PaviaU, Houston18, and TG (−0.55 dB on Chikusei, whose loss plateaus earliest); on PaviaU it stops at iteration 2800, saving 30% of the adaptation time.

V. CONCLUSION

This article presented AnyFusion, a universal unsupervised HSI–MSI fusion framework designed around the principle of "train once, fuse anywhere." By representing every spectral band as a set element identified by its physical center wavelength, the proposed SST decouples the fusion network from any specific sensor configuration: spectral set attention guided by a continuous $\Delta\lambda$ bias and an SRF-overlap routing prior handles arbitrary band sets, and a wavelength-query decoder synthesizes arbitrary output bands at arbitrary spatial scales. A per-scene degradation stage jointly learns an implicit neural PSF and a wavelength-conditional SRF from the observations alone, and was shown to reach the fusion accuracy of oracle degradation operators without knowledge of the kernel or of the sampling geometry. Built on this band-agnostic foundation, a cross-sensor universal pretraining stage, driven purely by unsupervised consistency losses, yields a single checkpoint that adapts rapidly to unseen sensors. Experiments on four benchmarks demonstrated the best mean PSNR among the compared unsupervised methods on three of the four datasets, pretraining gains of + 0.62 to + 2.77 dB with seed variance reduced by up to 26$\times$, and 2.0–3.3$\times$ cheaper adaptation at equal quality. The architecture accommodates arbitrary scales, band subsets, wavelength ranges, and SRFs without modification, and pretrained transfer holds across scales and SRFs: the pretraining gain remains positive at every adaptation scale from 2$\times$ to 6$\times$ despite pretraining only at 4$\times$, and reaches +2.40 dB in three-run mean under an SRF unseen during pretraining.

Several limitations point to future work. First, like other consistency-driven unsupervised methods, AnyFusion degrades under severe observation noise, since the consistency losses propagate noise into the reconstruction; Table XIV shows this to be a paradigm-level property (each architecture is most exposed to noise on the observation it anchors its spectral evidence to), and noise-aware degradation modeling is a natural extension. Second, the wavelength-coverage boundary is physical: target bands with no spectral overlap with any input band cannot be recovered, as the cross-band

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

experiments quantify. Third, the per-iteration cost of the band-token representation remains the highest among the compared methods: as the profiling in Section IV-H shows, the bottleneck is not arithmetic but the memory traffic of streaming per-band tokens at full spatial resolution, which suggests concrete remedies: processing tokens at a reduced spatial resolution with learned upsampling, grouping spectrally adjacent bands into shared tokens to shrink the set size N, and distilling the adapted model into a compact feed-forward student for repeated deployment on the same sensor. Because pretraining already cuts the iteration budget by 2.0–3.3$\times$ without touching the per-step cost, these two levers are complementary and multiplicative. Fourth, our evidence for sensor universality is bounded by the data: the pool contains four VNIR sensors simulated with one SRF (S2B) and a spectrally invariant Gaussian PSF. Transfer to another SRF, to non-Gaussian blur, and to a real spaceborne pair is positive, but a larger pool with more SRFs, SWIR sensors, and real degradations, including spectrally varying PSFs such as those of the separate VNIR and SWIR detectors of AHSI, is needed to substantiate it more broadly. Likewise, observation consistency and QNR are necessary rather than sufficient indicators of fusion quality on the preregistered ZY1-02D pair; robustness to spatially varying misregistration and task-based validation against independent ground truth, e.g., land-cover classification, are needed for operational deployment. Finally, AnyFusion has not been evaluated specifically on wetland or forest scenes, whose fine-scale heterogeneity and vertical structure challenge purely spectral–spatial fusion; integrating LiDAR or 3-D point clouds [4], [82] and validating fused products through vegetation retrieval [46] are promising extensions.

ACKNOWLEDGMENT

The authors would like to thank Prof. Paolo Gamba from the University of Pavia for providing the ROSIS-3 PaviaU data; the National Center for Airborne Laser Mapping (NCALM), the HSI Analysis Laboratory at the University of Houston, and the IEEE GRSS Image Analysis and Data Fusion Technical Committee for acquiring and providing the CASI Houston data used in the 2018 IEEE GRSS Data Fusion Contest; Prof. Naoto Yokoya and Prof. Akira Iwasaki from the Space Application Laboratory, the University of Tokyo, for providing the Chikusei hyperspectral data; and China Manned Space Engineering Office and the Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences, for providing the TG-1 hyperspectral data.

REFERENCES

[1] B. G. Ram, P. Oduor, C. Igathinathane, K. Howatt, and X. Sun, "A systematic review of hyperspectral imaging in precision agriculture: Analysis of its current state and future prospects," *Comput. Electron. Agricult.*, vol. 222, Jul. 2024, Art. no. 109037.

[2] S. Asadzadeh, X. Zhou, and S. Chabrillat, "Assessment of the space-borne EnMAP hyperspectral data for alteration mineral mapping: A case study of the reko diq porphyry cu au deposit, Pakistan," *Remote Sens. Environ.*, vol. 314, Dec. 2024, Art. no. 114389.

[3] Y. Ma, Y. Zhao, J. Im, Y. Zhao, and Z. Zhen, "A deep-learning-based tree species classification for natural secondary forests using unmanned aerial vehicle hyperspectral images and LiDAR," *Ecological Indicators*, vol. 159, Feb. 2024, Art. no. 111608.

[4] B. Fu et al., "Quantifying vegetation species functional traits along hydrologic gradients in Karst wetland based on 3D mapping with UAV hyperspectral point cloud," *Remote Sens. Environ.*, vol. 307, Jun. 2024, Art. no. 114160.

[5] Y. Liu, K. Jiang, W. Xie, J. Zhang, Y. Li, and L. Fang, "Hyperspectral anomaly detection with self-supervised anomaly prior," *Neural Netw.*, vol. 187, Jul. 2025, Art. no. 107294.

[6] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 2, pp. 29–56, Jun. 2017.

[7] R. Dian, S. Li, B. Sun, and A. Guo, "Recent advances and new guidelines on hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 69, pp. 40–51, May 2021.

[8] S. Luo, Y. Qian, L. Bai, Y. Fan, Y. Wang, and W. Kong, "Deep learning-based hyperspectral and multispectral fusion techniques: Review, optimization, and perspectives," *Inf. Fusion*, vol. 124, Dec. 2025, Art. no. 103291.

[9] H.-F. Yan, Y.-Q. Zhao, J. C.-W. Chan, S. G. Kong, N. Ei-Bendary, and M. Reda, "Hyperspectral and multispectral image fusion: When model-driven meet data-driven strategies," *Inf. Fusion*, vol. 116, Apr. 2025, Art. no. 102803.

[10] R. B. Gomez, A. Jazaeri, and M. Kafatos, "Wavelet-based hyperspectral and multispectral image fusion," in *Geo-Spatial Image Data Exploitation II*, vol. 4383. Bellingham, WA, USA: SPIE, 2001, pp. 36–42.

[11] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS +Pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[12] M. Selva, B. Aiazzi, F. Butera, L. Chiarantini, and S. Baronti, "Hyper-sharpening: A first approach on SIM-GA data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3008–3024, Jun. 2015.

[13] M. T. Eismann and R. C. Hardie, "Hyperspectral resolution enhancement using high-resolution multispectral imagery with arbitrary response functions," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 455–465, Mar. 2005.

[14] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Bayesian fusion of multi-band images," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 6, pp. 1117–1127, Sep. 2015.

[15] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[16] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[17] W. Dong et al., "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.

[18] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[19] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[20] W. Wan, W. Guo, H. Huang, and J. Liu, "Nonnegative and nonlocal sparse tensor factorization-based hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8384–8394, Dec. 2020.

[21] Y. Bu et al., "Hyperspectral and multispectral image fusion via graph Laplacian-guided coupled tensor decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 648–662, Jan. 2021.

[22] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.

[23] Q. Xie, M. Zhou, Q. Zhao, D. Meng, W. Zuo, and Z. Xu, "Multispectral and hyperspectral image fusion by MS/HS fusion net," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1585–1594.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

[24] J.-F. Hu, T.-Z. Huang, L.-J. Deng, T.-X. Jiang, G. Vivone, and J. Chanussot, "Hyperspectral image super-resolution via deep spatiospectral attention convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 7251–7265, Dec. 2022.

[25] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial–spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5953–5965, Jul. 2021.

[26] L. Sun et al., "MDC-FusFormer: Multiscale deep cross-fusion transformer network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528914.

[27] J. Fang, J. Yang, A. Khader, and L. Xiao, "MIMO-SST: Multi-input multi-output spatial–spectral transformer for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510020.

[28] C. Zhu, S. Deng, Y. Zhou, L.-J. Deng, and Q. Wu, "QIS-GAN: A lightweight adversarial network with quadtree implicit sampling for multispectral and hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531115.

[29] S. Peng, X. Zhu, H. Deng, L.-J. Deng, and Z. Lei, "FusionMamba: Efficient remote sensing image fusion with state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5410216.

[30] C. Wu, D. Wang, Y. Bai, H. Mao, Y. Li, and Q. Shen, "HSR-diff: Hyperspectral image super-resolution via conditional diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 7060–7070.

[31] Y.-J. Liang, Z. Cao, S. Deng, H.-X. Dou, and L.-J. Deng, "Fourier-enhanced implicit neural fusion network for multispectral and hyperspectral image fusion," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 63441–63465.

[32] Y. Qu, H. Qi, and C. Kwan, "Unsupervised sparse Dirichlet-net for hyperspectral image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2511–2520.

[33] K. Zheng et al., "Coupled convolutional neural network with adaptive response function learning for unsupervised hyperspectral super resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2487–2502, Mar. 2021.

[34] J. Yao, D. Hong, J. Chanussot, D. Meng, X. Zhu, and Z. Xu, "Cross-attention in coupled unmixing nets for unsupervised hyperspectral super-resolution," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2020, pp. 208–224.

[35] L. Zhang, J. Nie, W. Wei, Y. Li, and Y. Zhang, "Deep blind hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2388–2400, Jun. 2021.

[36] J. Liu, Z. Wu, L. Xiao, and X.-J. Wu, "Model inspired autoencoder for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522412.

[37] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[38] J. Li, K. Zheng, W. Liu, Z. Li, H. Yu, and L. Ni, "Model-guided coarse-to-fine fusion network for unsupervised hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[39] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Apr. 2023, Art. no. 5509417.

[40] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[41] H. Yu, Z. Ling, K. Zheng, L. Gao, J. Li, and J. Chanussot, "Unsupervised hyperspectral and multispectral image fusion with deep spectral–spatial collaborative constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534114.

[42] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[43] Y.-J. Liang, Z. Cao, L.-J. Deng, Y. Yang, and M. Zhang, "Hyperspectral image fusion with spectral-band and fusion-scale agnosticism," in *Proc. 43rd Int. Conf. Mach. Learn.*, 2026, pp. 1–21. [Online]. Available: https://icml.cc/virtual/2026/poster/65192

[44] Y. Zhang, "Wavelet-based Bayesian fusion of multispectral and hyperspectral images using Gaussian scale mixture model," *Int. J. Image Data Fusion*, vol. 3, no. 1, pp. 23–37, Mar. 2012.

[45] X. Cao, X. Wang, X. Dun, Y. Lian, X. Cheng, and X. Hao, "Cross-domain-aware deep unfolding transformer for hyperspectral image super-resolution," *Pattern Recognit.*, vol. 172, Apr. 2026, Art. no. 112374.

[46] B. Fu, X. Deng, H. Kuang, Z. Wang, and D. Fan, "GFHMP: Gradual fusion framework of hyperspectral, multispectral, and panchromatic images using a novel spatial–spectral cross-fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5528517.

[47] Z. Wang, B. Chen, R. Lu, H. Zhang, H. Liu, and P. K. Varshney, "FusionNet: An unsupervised convolutional variational network for hyperspectral and multispectral image fusion," *IEEE Trans. Image Process.*, vol. 29, pp. 7565–7577, 2020.

[48] J. Li, K. Zheng, J. Yao, L. Gao, and D. Hong, "Deep unsupervised blind hyperspectral and multispectral data fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[49] X. Cao, Y. Lian, K. Wang, C. Ma, and X. Xu, "Unsupervised hybrid network of transformer and CNN for blind hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5507615.

[50] H. Wang, Y. Xu, Z. Wu, and Z. Wei, "Unsupervised hyperspectral and multispectral image blind fusion based on deep Tucker decomposition network with spatial–spectral manifold learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 12721–12735, Jul. 2025.

[51] Y. Fang, Y. Liu, C.-Y. Chi, Z. Long, and C. Zhu, "CS2DIPs: Unsupervised HSI super-resolution using coupled spatial and spectral DIPs," *IEEE Trans. Image Process.*, vol. 33, pp. 3090–3101, 2024.

[52] J. Liu, Z. Wu, and L. Xiao, "A spectral diffusion prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528613.

[53] K. Li, L. Van Gool, and D. Dai, "Test-time training for hyperspectral image super-resolution," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 9, pp. 7231–7242, Sep. 2025.

[54] X. Qi et al., "A multistage semi-supervised network for hyperspectral super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5510017.

[55] Y. Cong et al., "SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 197–211.

[56] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[57] Z. Xiong et al., "Neural plasticity-inspired multimodal foundation model for Earth observation," 2024, *arXiv:2403.15356*.

[58] Y. Wang et al., "Towards a unified Copernicus foundation model for Earth vision," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 9888–9899.

[59] L. Waldmann et al., "Panopticon: Advancing any-sensor foundation models for Earth observation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2025, pp. 2195–2205.

[60] J. Prexl and M. Schmitt, "SenPa-MAE: Sensor parameter aware masked autoencoder for multi-satellite self-supervised pretraining," in *Pattern Recognition–(DAGM GCPR) (Lecture Notes in Computer Science)*. Cham, Switzerland: Springer, 2025, pp. 317–331.

[61] G. Sumbul, C. Xu, E. Dalsasso, and D. Tuia, "SMARTIES: Spectrum-aware multi-sensor auto-encoder for remote sensing images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 5569–5578.

[62] K. Zheng, A. Chen, H. Yu, J. Li, and B. Zhang, "Unsupervised pretraining framework guided hyperspectral and multispectral image fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[63] J. Jia, H. Yu, C. Wang, K. Zheng, J. Li, and J. Hu, "Spectral–spatial collaborative pretraining framework with multiconstraint cooperation for hyperspectral–multispectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 11610–11622, 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.

[64] Z. Xiong, Z. Shi, H. Li, L. Wang, D. Liu, and F. Wu, "HSCNN: CNN-based hyperspectral image recovery from spectrally undersampled projections," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 518–525.

[65] J. Li, C. Wu, R. Song, Y. Li, and F. Liu, "Adaptive weighted attention network with camera spectral sensitivity prior for spectral reconstruction from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 462–463.

[66] Y. Cai et al., "MST++: Multi-stage spectral-wise transformer for efficient spectral reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 744–754.

[67] V. Sitzmann, J. Martel, A. Bergman, D. Lindell, and G. Wetzstein, "Implicit neural representations with periodic activation functions," in *Proc. NIPS*, 2020, pp. 7462–7473.

[68] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," in *Proc. Eur. Conf. Comput. Vis*. Glasgow, U.K.: Springer, Aug. 2020, pp. 405–421.

[69] M. Tancik et al., "Fourier features let networks learn high frequency functions in low dimensional domains," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 7537–7547.

[70] Y. Chen, S. Liu, and X. Wang, "Learning continuous image representation with local implicit image function," in *Proc. IEEE/CVF Conf. Comput. Vis. pattern Recognit.*, Jul. 2021, pp. 8628–8638.

[71] G. Mai et al., "SSIF: Learning continuous image representation for spatial–spectral super-resolution," 2023, *arXiv:2310.00413*.

[72] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inform. Process. Syst. (NIPS)*, 2017, pp. 5998–6008.

[73] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Poczos, R. R. Salakhutdinov, and A. J. Smola, "Deep sets," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 3391–3401.

[74] J. Lee, Y. Lee, J. Kim, A. R. Kosiorek, S. Choi, and Y. W. Teho, "Set transformer: A framework for attention-based permutation-invariant neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 3744–3753.

[75] E. Perez, F. Strub, H. De Vries, V. Dumoulin, and A. Courville, "FiLM: Visual reasoning with a general conditioning layer," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, 2018, pp. 1–13.

[76] P. Gamba, "A collection of data for urban area characterization," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Anchorage, AK, USA, Sep. 2004, pp. 69–72.

[77] Y. Xu et al., "Advanced multi-sensor optical remote sensing for urban land use and land cover classification: Outcome of the 2018 IEEE GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1709–1724, Jun. 2019.

[78] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over Chikusei," Space Appl. Lab., Univ. Tokyo, Tokyo, Japan, Tech. Rep. SAL-2016-05-27, May 2016.

[79] X. Li, T. Wu, K. Liu, Y. Li, and L. Zhang, "Evaluation of the Chinese fine spatial resolution hyperspectral satellite TianGong-1 in urban land-cover classification," *Remote Sens.*, vol. 8, no. 5, p. 438, May 2016.

[80] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[81] L. Alparone, B. Aiazzi, S. Baronti, A. Garzelli, F. Nencini, and M. Selva, "Multispectral and panchromatic data fusion assessment without reference," *Photogramm. Eng. Remote Sens.*, vol. 74, no. 2, pp. 193–200, Feb. 2008.

[82] L. Deng et al., "Comparison of 2D and 3D vegetation species mapping in three natural scenarios using UAV-LiDAR point clouds and improved deep learning methods," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 125, Dec. 2023, Art. no. 103588.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:21:43 UTC from IEEE Xplore. Restrictions apply.